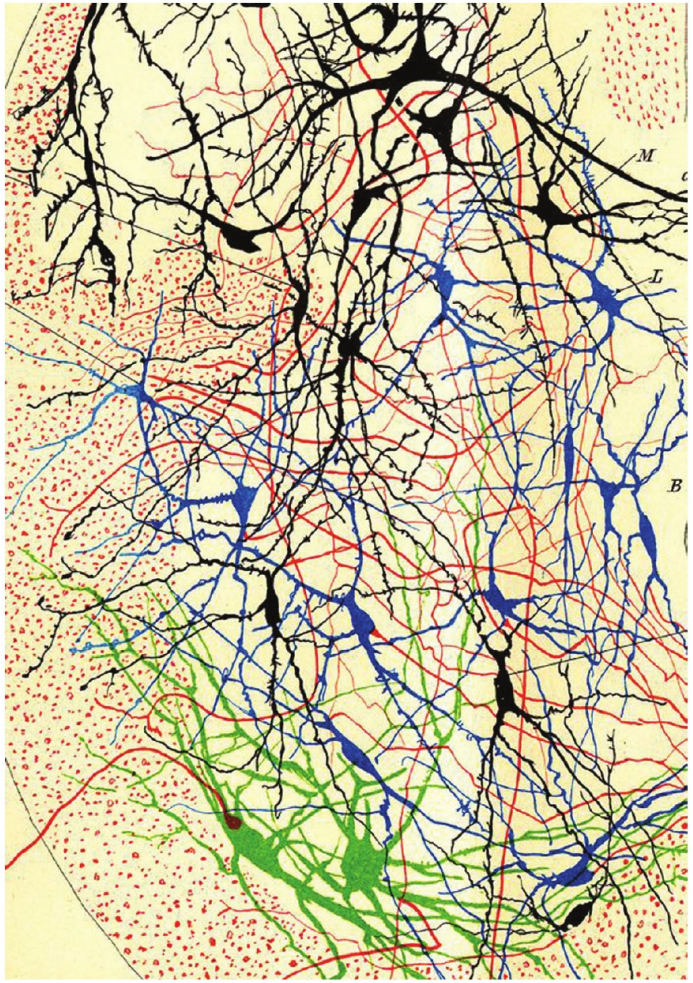


Science Source

Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal’s drawing of Golgi-stained neurons.

BEHAVIORAL NEUROSCIENCE 392

Foundations and Frontiers

From Brain to Behavior: Foundations and Frontiers of Behavioral Neuroscience

Fall 2026

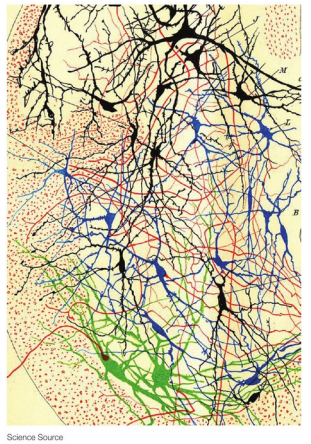
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Brain activity produces perception and action

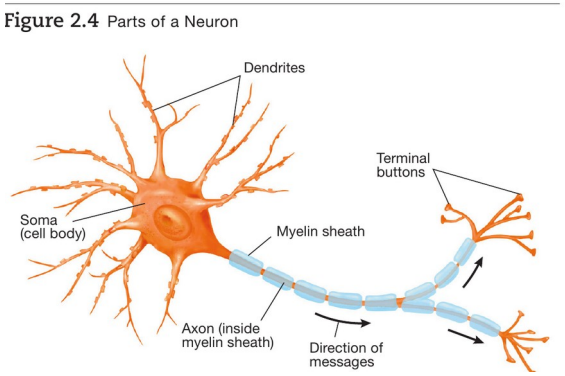
Behavioral neuroscience studies how the nervous system produces perception, movement, memory, and emotion. A **neuron** is a nerve cell. Neurons receive signals, change their electrical activity, and send signals to other cells. Incoming signals can increase or decrease a cell's tendency to fire.

Sensory systems detect events outside and inside the body. Motor systems organize actions such as reaching, walking, or speaking. Memory allows an earlier experience to change a later response. For example, seeing food can guide the direction of a reach. Hunger changes whether the person starts that action, and a remembered location changes where the person looks first.

An experiment can measure both the action and the activity of the cells involved. The scientist asks what the animal does, which cells are active, and how changing those cells changes the action. Recording identifies when activity occurs during the action. Changing the activity tests whether the action also changes, giving a more specific explanation than describing the movement alone.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



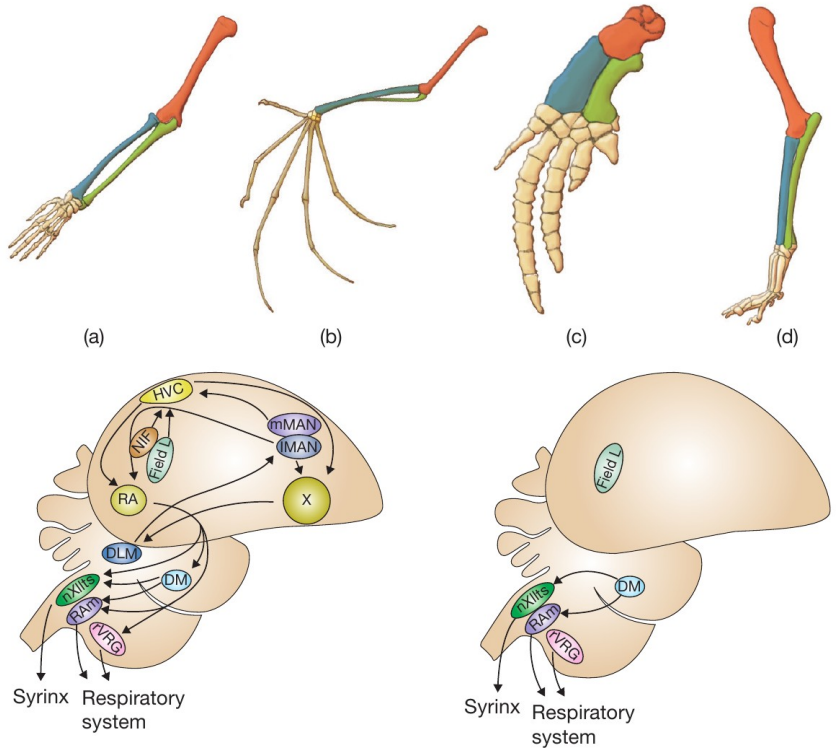
Carlson and Birkett (2022), Fig. 2.4, p. 40. Labeled cell body, dendrites, axon, and terminals.

Generalization identifies recurring behavioral rules

Generalization identifies a rule that applies to several observations. Comparing the same behavior in different situations reveals which conditions reliably change it. A useful comparison changes one circumstance while keeping the action measurable. The researcher can then describe which conditions increase, decrease, or leave the action unchanged.

A mouse may build a nest when its surroundings are cold. Pregnancy can also increase nest building, even when cold is not the immediate trigger. The animal can perform a similar construction sequence for different reasons. Cooling and pregnancy therefore provide two ways to investigate when the same observable movements begin.

The movements are similar, but the conditions that start them differ. Describing those conditions gives a behavioral rule. Investigating the signals that trigger the movements explains how the rule arises. The behavior can be recognized across situations without assuming its trigger is identical. Separating the visible action from the signals initiating it makes the explanation more precise.



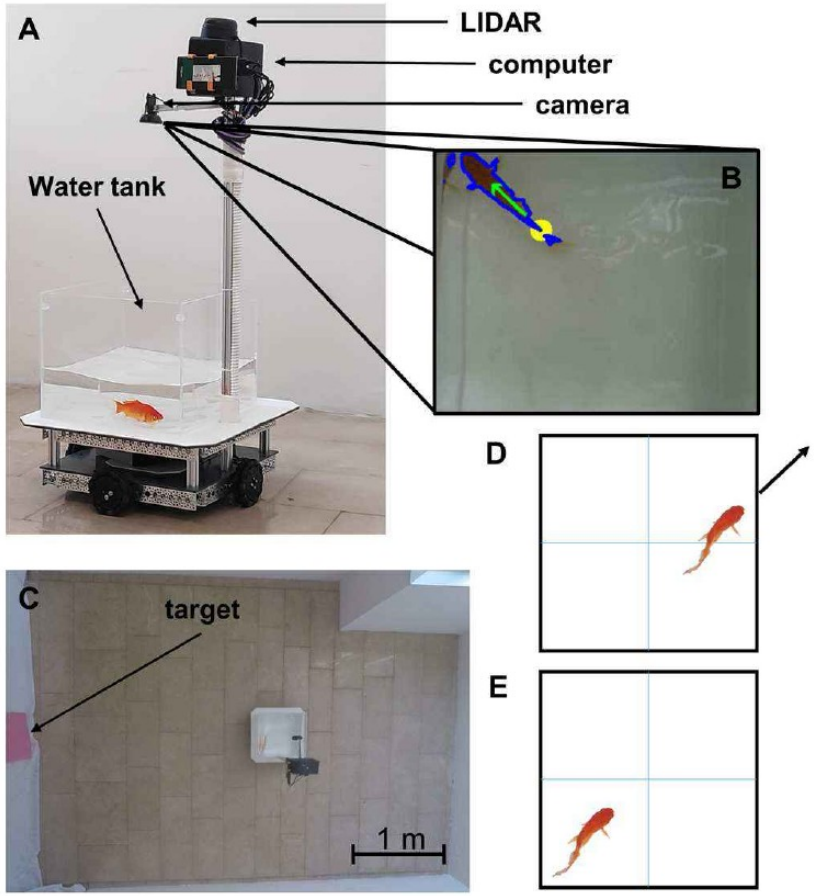
Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.

Goldfish drive toward a terrestrial visual target

Givon and colleagues placed a goldfish tank on a wheeled platform. The **fish-operated vehicle**, or FOV, moved according to the fish's location and the direction it faced. The device translated an action within water into movement across land. The fish did not leave the tank; it experienced the room changing position around its enclosure.

A pink board marked the destination in a 3 × 4 metre room. Reaching it produced a food pellet. The goldfish remained in water while its swimming actions moved the tank across the floor. The visible board supplied a destination outside the water. The food consequence followed arrival, allowing the fish to experience which swimming actions successfully moved the device toward the goal.

With training, arrivals became more frequent and routes more direct. The fish learned to use an unfamiliar movement system to approach a visual goal outside its tank. The improvement concerned selecting effective actions within the tank. Successful navigation required both recognizing the destination and using the device's control rules to approach it.



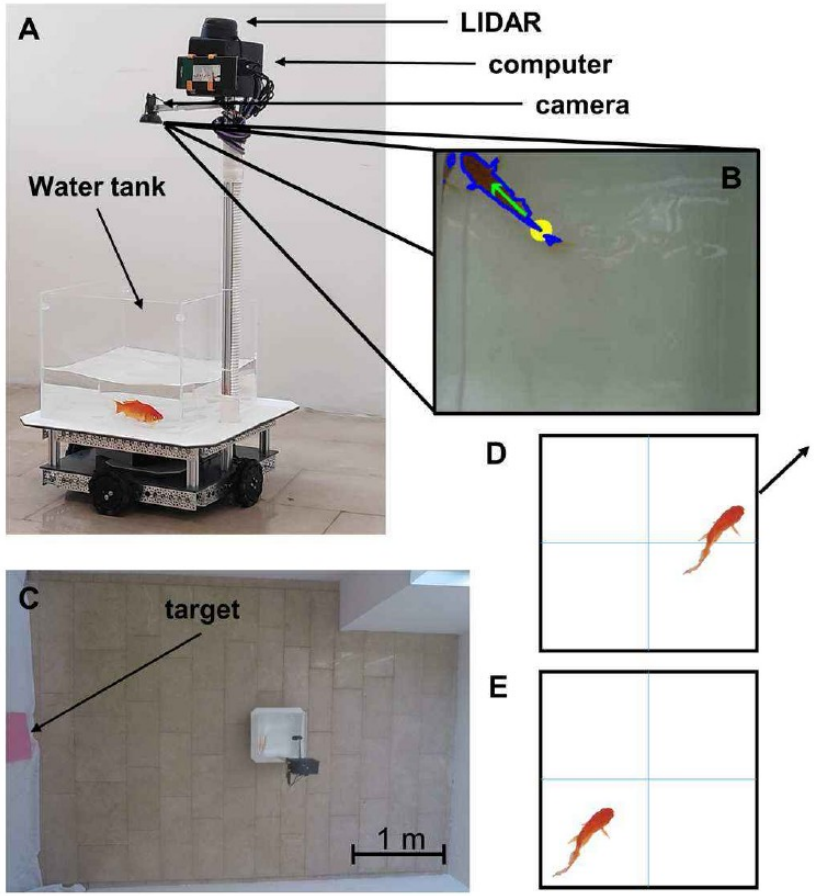
Givon et al. (2022), Fig. 1(A–E). The fish-operated vehicle and its control mapping.

Fish position determines vehicle movement

A camera above the tank measured the goldfish's position and orientation. Software identified its body outline and estimated the direction it faced. Position and orientation are separate measurements. A fish can occupy the same location while facing different directions, so both are needed to determine an appropriate movement command.

Near a tank boundary, facing outward could command the wheels to move. Swimming to a new location inside the tank therefore changed the platform's direction through the room. The camera observation became an input to the motor controller. The wheels then displaced the tank, changing the distant scene that the fish could use to guide its next action.

The researchers recorded both fish position and vehicle position. The first measured the controlling action; the second measured the resulting route. These records made changes during training measurable. A route through the room measures the combined result of many controlling actions. Recording the two positions separately makes it possible to examine how a particular input affected progress.



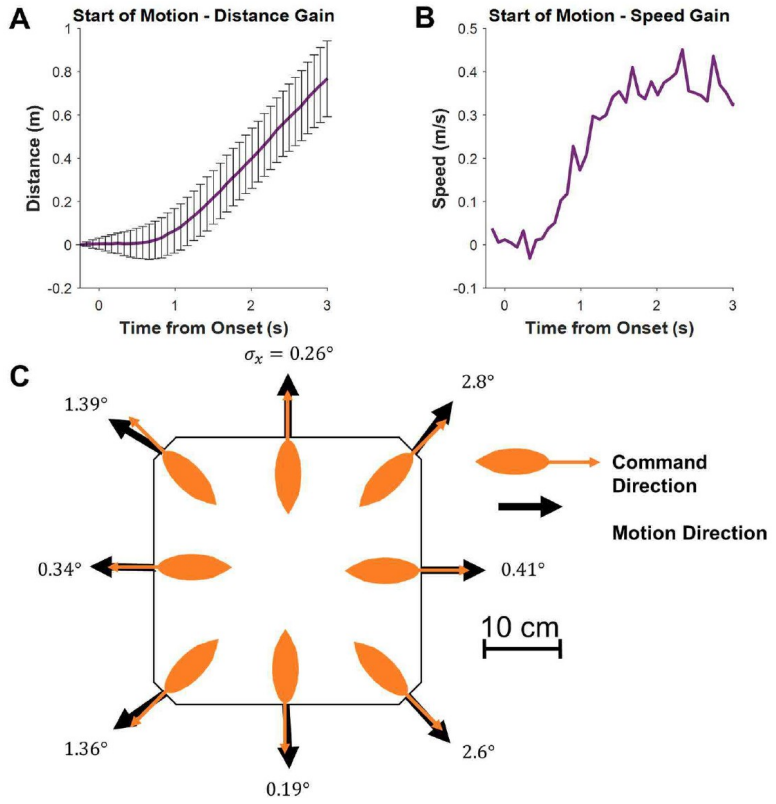
Givon et al. (2022), Fig. 1(A–E). The fish-operated vehicle and its control mapping.

Orientation and tank position control the wheels

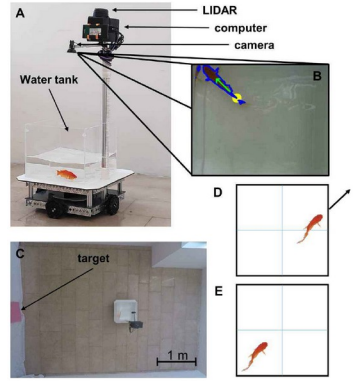
The vehicle moved when the fish approached a boundary and faced outward. Facing inward or staying near the tank center could leave the platform stationary. A movement command required the specified posture near the boundary. This rule made some swimming actions effective for controlling the platform while leaving other actions without forward displacement.

Wheel movement developed over about one second after a command. The fish therefore experienced a delay between changing position and seeing the tank move through the room. The visual consequence followed the controlling action after a mechanical delay. The animal had to keep or change its action while the platform's response was still developing.

A laser-ranging system prevented movement within 20 centimetres of an obstacle. The fish's actions, the mechanical delay, and automatic collision prevention all shaped the routes available. Automatic stopping could interrupt an otherwise valid movement command. The available path therefore depended on both the fish's decisions and the device's obstacle-detection rule.



Givon et al. (2022), Fig. 2(A–C). Motor delay and directional response of the vehicle.



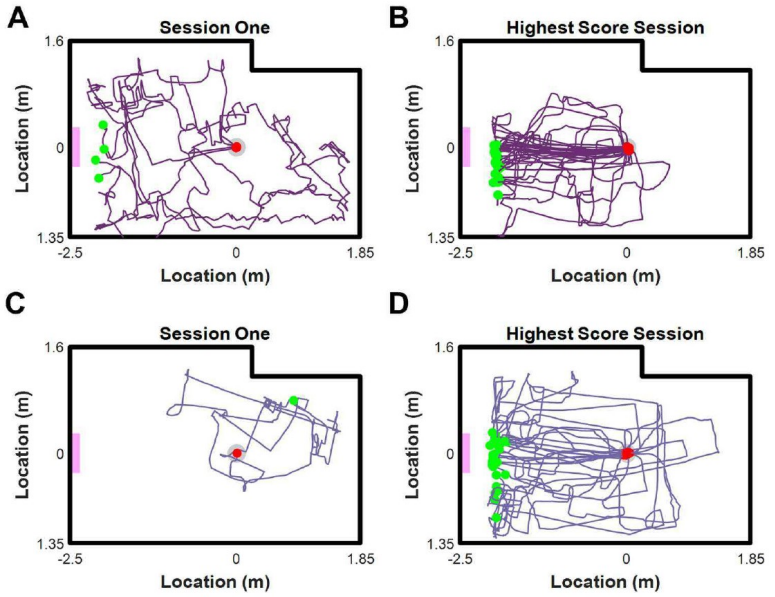
Givon et al. (2022), Fig. 1(A–E). The fish-operated vehicle and its control mapping.

Rewarded training produces more direct routes

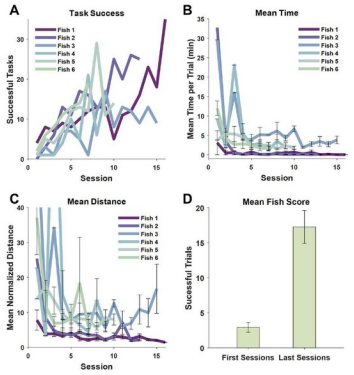
Operant learning changes behavior through its consequences. In the vehicle task, arrival at the pink board produced food. The fish could try again after the tank returned to the start. The food consequence occurred after reaching a defined location. Repeated trials gave the fish opportunities to change its actions according to whether those actions produced arrival.

Early routes wandered and often missed the target. Later routes became more direct, with more successful arrivals and shorter journeys. A shorter journey could involve fewer detours or better choices of direction. Measuring the route as well as success distinguishes reaching a goal from reaching it efficiently.

Reward followed successful control of the vehicle. Across repeated attempts, the fish changed its movements so the tank reached the goal more efficiently. The learned action concerns operating the device under its control rules. As the fish gains experience, the same environmental goal can be reached with a different sequence of swimming movements.



Givon et al. (2022), Fig. 3(A–D). Routes become more direct with training.



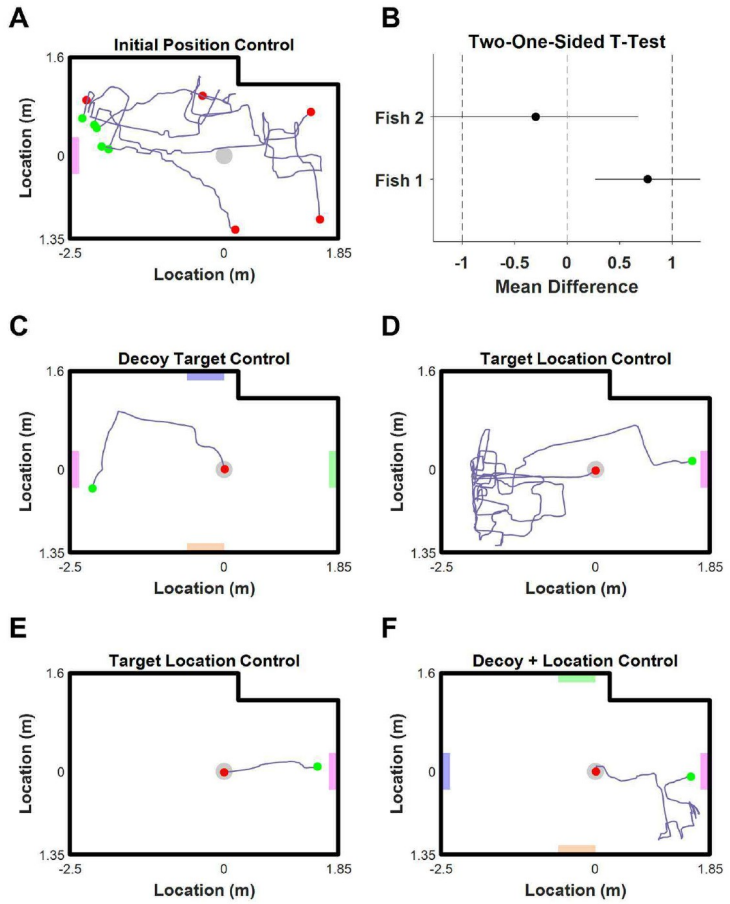
Givon et al. (2022), Fig. 4(A–D). Training improves target arrivals, travel time, and path length.

Changed starting positions test route flexibility

Givon and colleagues changed where the vehicle started. The visual target remained the destination, but the fish needed a different route to reach it. A different start changes which directions initially lead toward the board. Keeping the destination fixed therefore preserves the goal while changing the movements needed for arrival.

Fish could approach the target from these new starting positions. Their actions were therefore not confined to repeating one trained path through the room. A successful new route requires using information about the present situation. The fish cannot arrive by assuming that every trial begins at the previously trained position.

A visible goal can guide movement when the starting point changes. The fish must select movements using its current position rather than only reproduce a fixed sequence. Flexibility can be measured from the resulting trajectory and target arrivals. These outcomes describe how the animal adjusts its controlling actions when the starting conditions change.



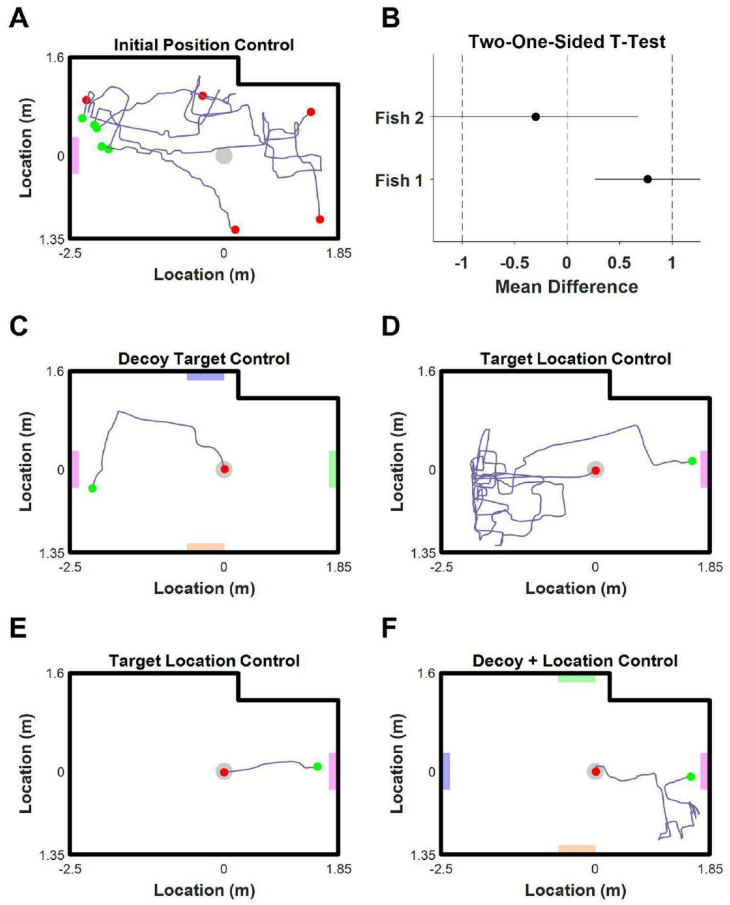
Givon et al. (2022), Fig. 5(A–F). Changing starts, decoys, and relocated targets tests navigation.

Decoys and target relocation test visual guidance

The researchers added visual decoys and relocated the target. These changes tested which visual information guided the vehicle's movement. A decoy can resemble part of the trained visual scene without supplying the same reward. Its presence changes the available cues while leaving the navigation task measurable.

Fish continued to approach the rewarded target under changed conditions. The experiments varied objects in the room while recording where the vehicle went. Relocating the destination changes which movements lead toward it. The fish must use the scene available on that trial rather than simply move toward the earlier location.

Navigation required detecting the destination and choosing actions that moved the tank toward it. Changes in the scene revealed how the fish used external visual information during the task. The platform's route records the result of those choices. Comparing routes across altered scenes identifies which visible cues continue to guide the animal toward the rewarded destination.



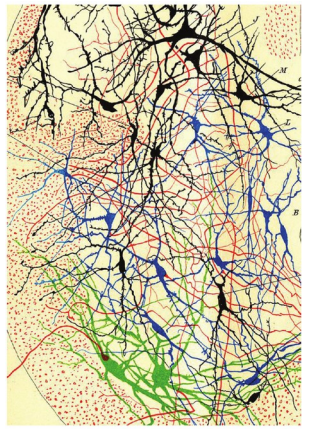
Givon et al. (2022), Fig. 5(A–F). Changing starts, decoys, and relocated targets tests navigation.

Reduction explains behavior through its parts

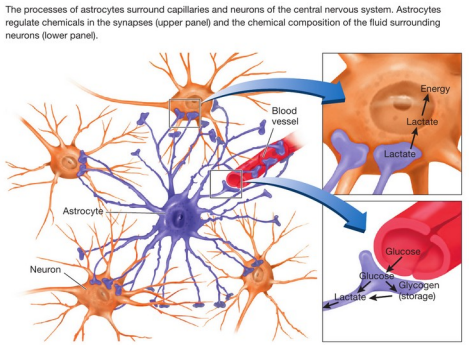
Reduction explains a complex event through its component processes. Nest building can be examined as sensory detection, internal signals, and movements that carry material into a nest. The whole action can be divided into collecting material, transporting it, and placing it. Each component requires sensory information and coordinated muscle activity at the appropriate time.

Temperature-sensitive systems detect cooling. Pregnancy alters **hormones**, chemical messengers released within the body. These signals act on neural systems that organize the animal's movements. A temperature-related input and a reproductive signal can act on systems that organize the same movements. Different internal conditions can therefore recruit a shared motor sequence.

Neurons also need a stable chemical environment. **Astrocytes**, star-shaped supporting cells, help regulate this environment and supply metabolic support. Explaining a behavior can therefore involve both neurons and the cells maintaining them. The surrounding fluid must contain suitable concentrations of ions and nutrients.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



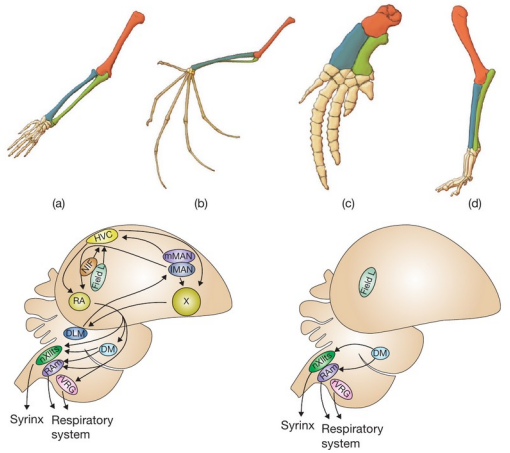
Carlson and Birkett (2022), Fig. 2.10, p. 44. Astrocytes, neurons, and metabolic support.

Behavior has a function and a biological mechanism

A behavior's **function** describes what it accomplishes. A nest helps maintain body temperature and shelters offspring. Its **mechanism** describes the processes that produce the construction movements. An outcome such as retaining heat is different from a description of the movements. Both explanations are needed to account for what the animal achieves and how it achieves it.

Sensory signals, hormones, and motor circuits determine when the animal gathers material and where it places it. The completed nest then changes the animal's surroundings. The animal must detect useful material, approach it, and coordinate carrying movements. Changing the sensory input or internal signal can change when this sequence begins.

The barreleye fish provides a sensory example. Its transparent head region and conspicuous eyes are visible in the footage. Studying an eye's structure asks how it gathers information; studying its use asks how that information guides behavior. A sensory structure determines which information is available to the nervous system.



Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.



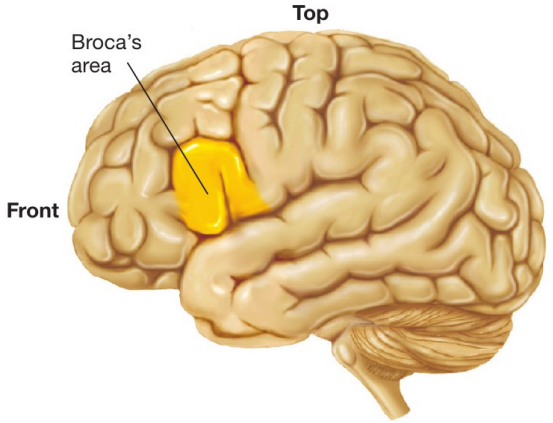
Visual context: MBARI, natural-history footage. Barreleye (Macropinna microstoma).

Brain and body systems coordinate perception and action

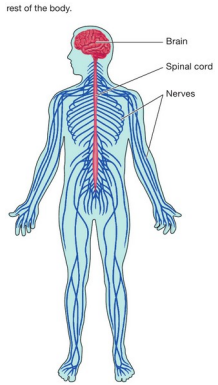
The **central nervous system** consists of the brain and spinal cord. **Peripheral nerves** carry information between this central system and the body. Sensory information enters through nerves, and motor commands return to muscles. The spinal cord also organizes some responses locally.

The **cerebral cortex** is the outer sheet of the cerebrum. Different cortical regions contribute to vision, movement, and language. These regions exchange signals as a person sees an object and reaches for it. A visual signal can reach systems that identify the object and systems that guide the hand.

Brain damage can interrupt a particular step in this process. A person may still see an object but have difficulty naming it or moving toward it. The affected behavior helps identify which brain functions have changed. The location of the injury helps distinguish a sensory problem from a motor or language problem.



Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca's area in the left hemisphere.



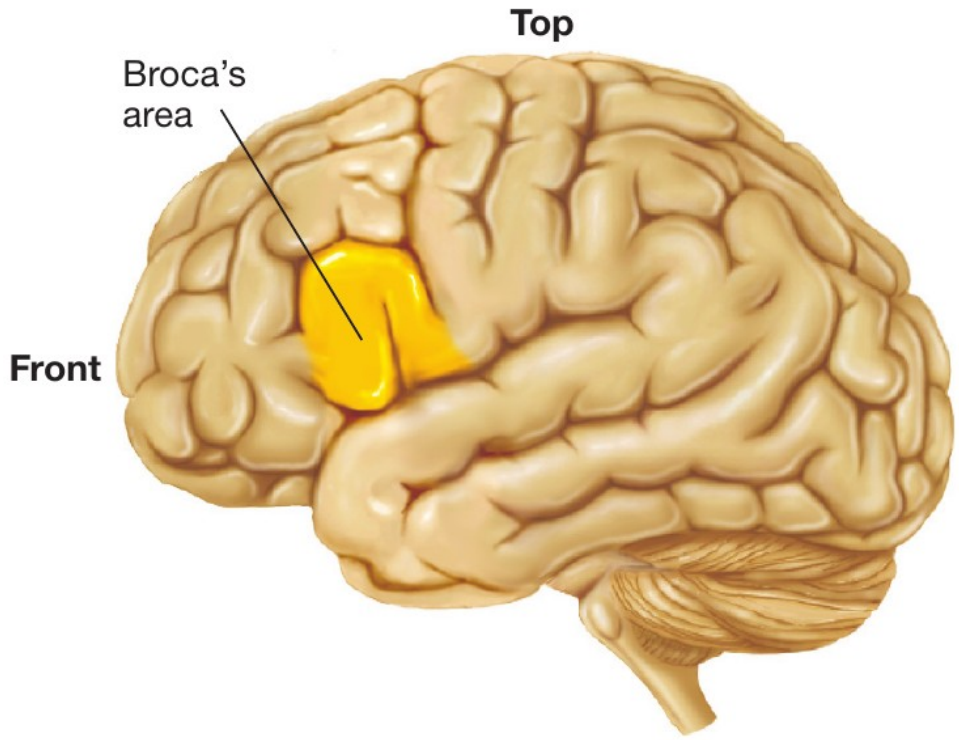
Carlson and Birkett (2022), Fig. 2.1, p. 38. Brain, spinal cord, and peripheral nerves.

Anatomical observations localized speech-related function

Paul Broca examined a person who had lost much of his ability to produce speech. After death, the brain contained damage in the left frontal region. A large loss of speech production occurred alongside a localized injury. Comparing the preserved abilities with the disrupted ability helped specify which part of language was affected.

Localization of function means that particular regions make particular contributions to behavior. **Broca's area** participates in processes needed for producing speech. Damage can interfere with converting intended words into spoken output. Producing a word requires retrieving its form and organizing the movements needed to say it.

Language also requires other regions for hearing, recognizing words, and retrieving information. Speaking depends on signals passing among these systems, not only on the activity of one location. A spoken exchange also requires hearing the other person and interpreting the words. These different operations explain why language can be disrupted in several distinct ways after injury.



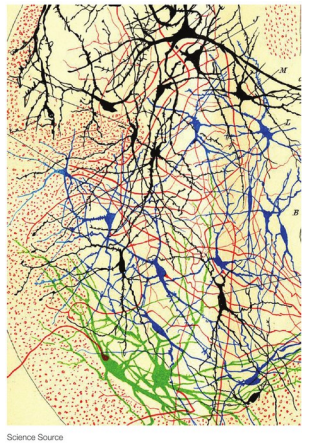
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca's area in the left hemisphere.

Staining revealed the cellular organization of the brain

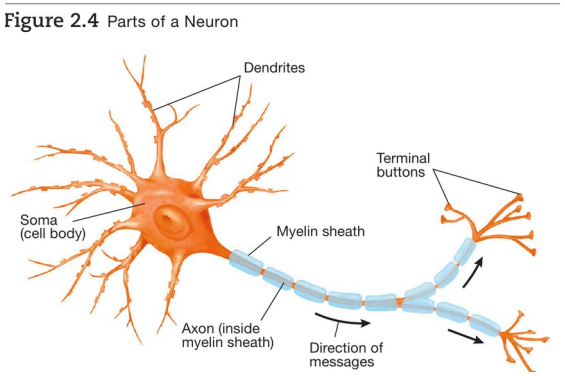
A stain makes selected structures visible under a microscope. The **Golgi method** colors a small fraction of neurons, revealing their cell bodies and branching processes. Coloring only some cells makes their branches easier to distinguish from neighboring tissue. A densely stained preparation would make individual processes more difficult to follow through the brain.

Santiago Ramón y Cajal examined these stained cells. He described neurons as distinct cells whose processes approach other cells. This cellular organization became central to explanations of neural signaling. Separate cells can communicate without forming one continuous cell.

Dendrites receive many incoming signals. An **axon** carries outgoing electrical signals toward terminals. The labeled neuron image identifies these structures, which perform different steps in communication. The cell body contains the nucleus and much of the machinery maintaining the neuron. Branching processes allow incoming signals and outgoing communication to occur at many locations.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



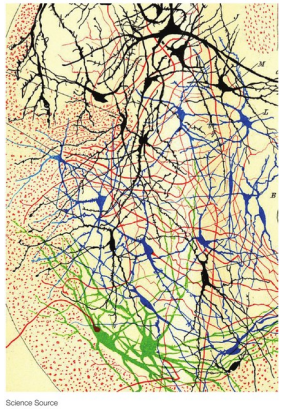
Carlson and Birkett (2022), Fig. 2.4, p. 40. Labeled cell body, dendrites, axon, and terminals.

Abnormal neural activity and recovery reveal plasticity

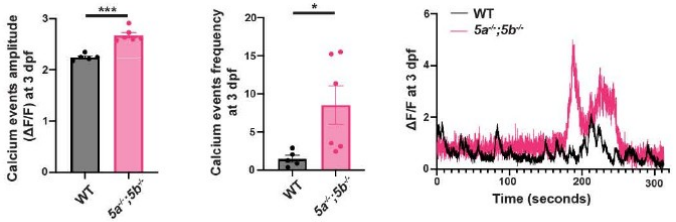
A **seizure** is a period of abnormal neural activity that can disrupt sensation, movement, or awareness. Rasmussen encephalitis causes brain inflammation and can produce repeated seizures in one cerebral hemisphere. A seizure can affect a small region or spread through a larger network. The affected neurons determine which sensations, movements, or changes in awareness occur during the episode.

A **hemisphere** is one of the cerebrum's two large sides. In severe childhood cases, disconnecting the affected side can reduce seizures. **Plasticity** allows remaining circuits to change as the child develops and practices skills. Remaining tissue must perform functions previously distributed across more of the brain.

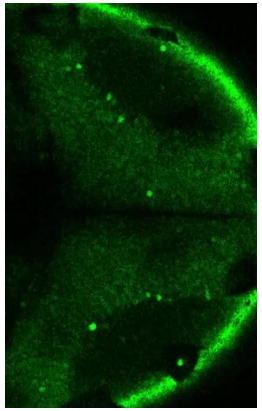
Dogra and colleagues recorded a zebrafish epilepsy model using a fluorescent calcium indicator. Bright changes mark calcium events in brain cells. The fish carries altered citrate-transport genes; its neural activity provides an experimental example of abnormal excitability. Measuring successive events reveals when individual cells become active within the recorded brain region.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



Dogra et al. (2025), Fig. 6A–C. Calcium-event measurements and a single-neuron trace in a zebrafish epilepsy model.



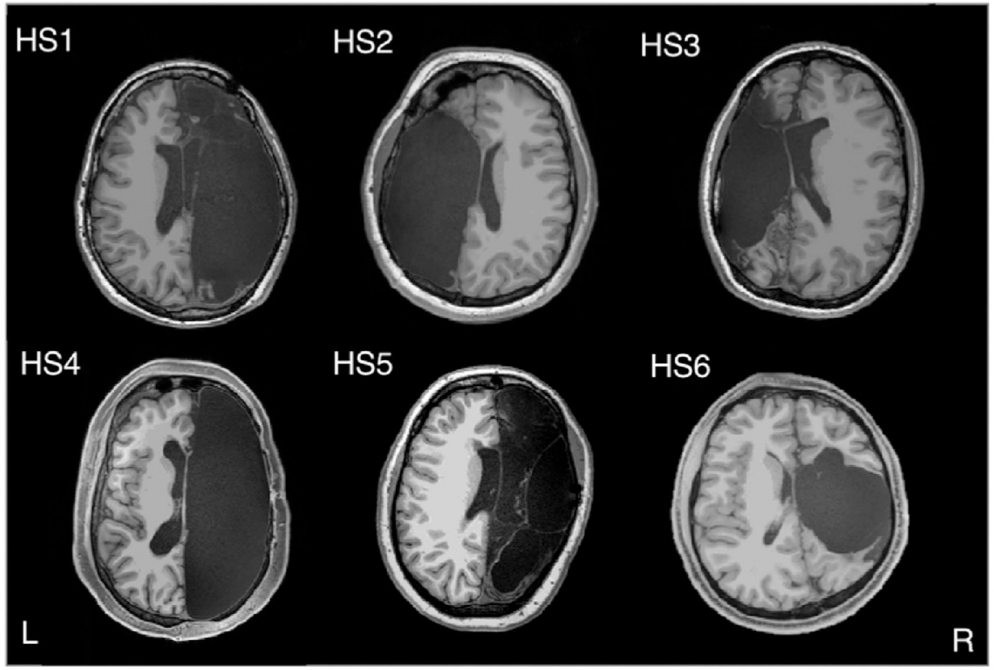
Dogra et al. (2025), S2 Movie. Midbrain calcium events in a zebrafish epilepsy model.

A single hemisphere retains organized brain networks

Kliemann and colleagues studied adults who had undergone **hemispherectomy**, removal or disconnection of one cerebral hemisphere, during childhood. Structural magnetic resonance imaging, or **MRI**, documented the remaining brain tissue. The operation profoundly changes available anatomy.

The adults retained recognizable networks for vision, movement, attention, and other functions. A **brain network** is a set of regions whose activity is organized together. A region is assigned to a network using the pattern of its activity. Regions with similar fluctuations can form a recognizable group even when the opposite hemisphere is absent.

The remaining hemisphere contained several distinct systems. Childhood brain reorganization preserved this division of work while changing how strongly different systems interacted. Reorganization does not mean every region performs every function equally. Distinct systems remain visible, while their interactions differ from the organization measured in people with two hemispheres.



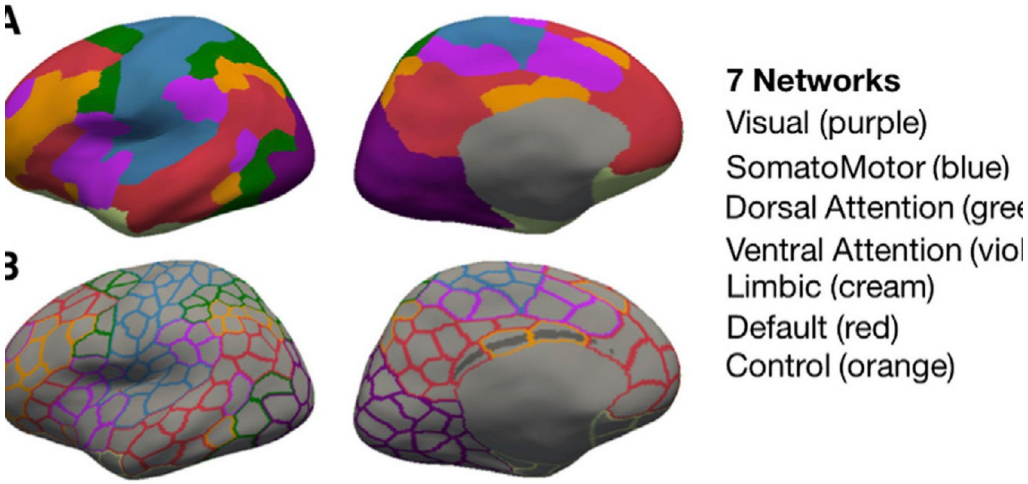
Kliemann et al. (2019), Fig. 1. Anatomy after childhood hemispherectomy.

Brain activity can be measured through its blood supply

Active brain tissue needs oxygen and nutrients. Local blood flow changes with neural activity. **Functional MRI**, or fMRI, measures a signal related to blood oxygenation across brain regions. The measured signal reflects blood oxygenation rather than individual action potentials. It is useful for comparing broad patterns across locations that cannot all be sampled with one electrode.

Functional connectivity describes how closely two regions' signals fluctuate together. Similar rises and falls indicate coordinated activity over the measured period. Researchers compare many regions to identify networks. Two signals can vary together even when their regions have different roles. Comparing many such pairs reveals which areas tend to participate in the same organized pattern.

Visual, movement-related, and attention-related areas form distinguishable groups. Comparing their fluctuations reveals how activity is organized within a hemisphere, including after childhood surgery. The map assigns regions to functional systems. The next measurement asks how closely their signals fluctuate within a system and between systems in the remaining brain.



Kliemann et al. (2019), Fig. 2A–B. Networks and regional parcellation.

Different brain networks interact more strongly after surgery

Kliemann and colleagues compared activity within a network with activity between different networks. Adults with childhood hemispherectomy retained strong coordination within their remaining brain networks. Within-network coordination describes how activity varies among areas belonging to the same system. This measurement can be compared with coordination between areas assigned to different systems.

Different networks also fluctuated together more strongly than in most participants with two hemispheres. Visual, motor, and attention-related systems remained distinguishable but interacted more extensively. A stronger interaction means that the measured signals are more similar over time. The change appears between functional systems rather than eliminating the distinction among those systems.

The authors proposed that greater interaction among networks helps describe the brain's reorganization after early surgery. The remaining tissue maintains several kinds of function within one hemisphere. This hypothesis concerns the organization of the surviving tissue. It directs further questions about how a single hemisphere performs functions that normally involve tissue on both sides.

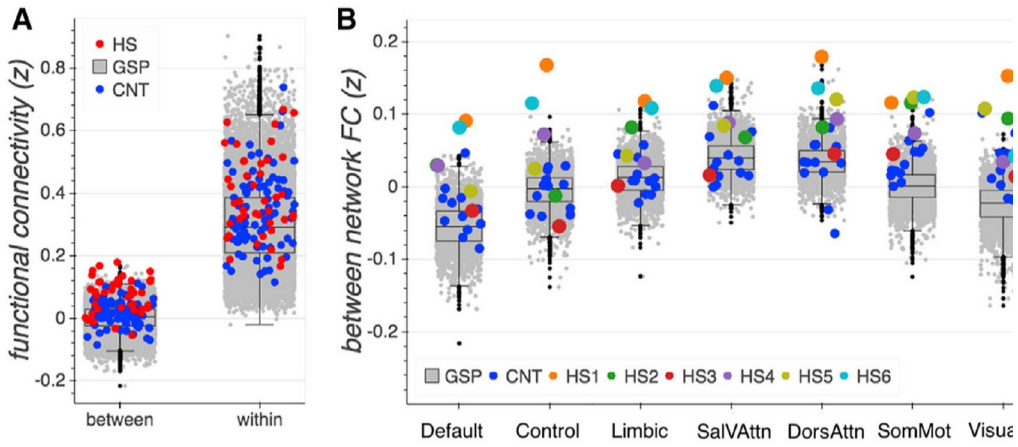


Figure 4. Functional Connectivity
(A) Between- and within-network functional connectivity averaged across networks per group (GSP, CNT, and HS) (seven data points, one for each network, per participant). CNT and HS showed similar within-network connectivity as compared to the large GSP sample, while overall between-network connectivity was notably stronger for HS participants. (B) High between-network connectivity was evident across all networks and in all but one

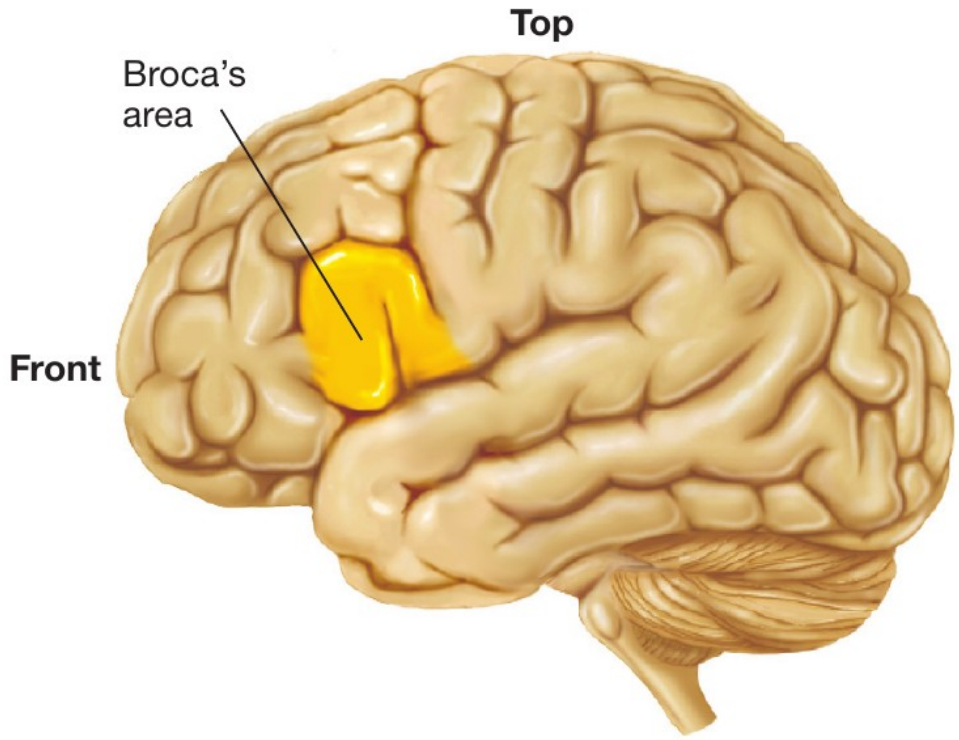
Kliemann et al. (2019), Fig. 4A–B. Within-network and between-network connectivity.

Biological explanations replaced early heart-centered ideas

Hippocrates associated thought and emotion with the brain. Aristotle assigned a major role to the heart and described the brain as a cooling organ. These accounts placed mental functions in different organs. The disagreement concerned the organ responsible for mental activity. An anatomical explanation must account for how information from a sense organ reaches the proposed processing structure.

Galen examined nervous systems in several animals. Nerves ran between sense organs, muscles, and the brain. Their paths made a brain-centered explanation of sensation and action anatomically plausible. Sensory nerves convey information toward the central nervous system. Motor nerves convey commands toward muscles, giving anatomical study routes to examine for different kinds of behavior.

Later investigators tested the effects of damaged nerves and brain tissue. A change in sensation or movement after damage helped identify the structure’s contribution to behavior. An interruption along a nerve can prevent a signal reaching its destination. Examining the lost function therefore identifies an important step in the pathway producing the normal response.



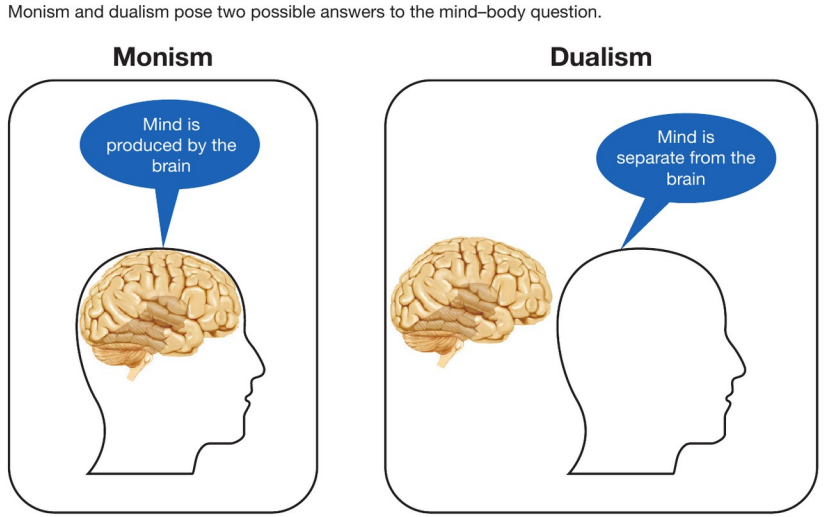
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca’s area in the left hemisphere.

Monism treats mental activity as brain function

Dualism treats mind and body as different kinds of entities. **Monism**, in a biological account, treats mental activity as a function of the physical brain. These positions offer different explanations of how thoughts and bodily actions occur. A biological account investigates the physical events accompanying the experience and the resulting response.

Descartes proposed mechanical explanations for some nervous-system responses. He also proposed that a nonphysical mind interacted with the body through the pineal gland. This was a historical proposal about mental activity. The proposal preceded modern measurements of neuronal activity.

Neuroscience measures physical events during perception, memory, and action. It also changes a biological process and measures the resulting behavior. These experiments examine how mental functions arise from the nervous system. Measuring an event identifies its timing and location.



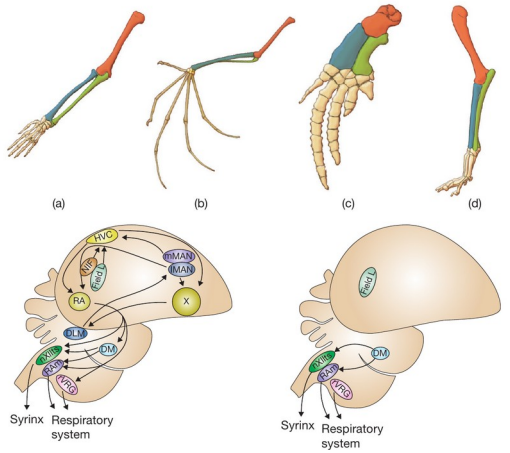
Carlson and Birkett (2022), Fig. 1.2, p. 19. Historical accounts of the mind–body relationship.

Selection changes inherited traits across generations

Natural selection changes inherited traits across generations. When a heritable trait affects survival or reproduction, individuals carrying it may leave different numbers of offspring. The relevant differences must be passed to offspring for their frequencies to change across generations. A temporary change within one animal is different from an inherited population change.

The next generation then contains a different distribution of traits. Selection acts on existing variation; individual animals do not acquire a useful inherited trait because they need it. Differences in reproduction determine which variants become more common. Environmental conditions affect those consequences, so a trait can have different effects in different surroundings.

The dragonfish footage provides a visible example of unusual body and light-producing structures. Explaining their evolution requires evidence about inheritance and reproductive consequences. Their appearance alone identifies traits to investigate. The first biological question is what these structures accomplish for the fish.



Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.



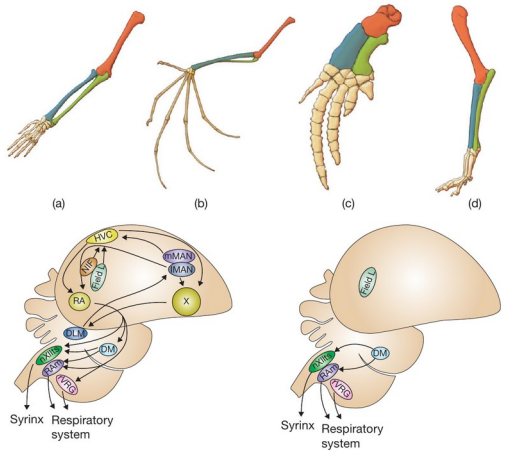
Visual context: MBARI, natural-history footage. Dragonfish (Idiacanthus antrostomus).

Functionalism relates traits to their biological roles

Functionalism asks what a trait accomplishes for the organism. A limb used for grasping and a limb used for flight perform different tasks even when they share an evolutionary origin. Shared ancestry can produce recognizable structural similarities despite different uses. Comparing the structures identifies both the inherited parts and the changes associated with their current activities.

An action can be described through both its outcome and the processes producing it. Feeding obtains food; sensory detection and motor activity determine how the animal captures or collects it. The outcome and the mechanism describe different parts of the explanation. The first concerns obtaining food; the second concerns detecting material and producing the movements needed to obtain it.

The vampire squid’s arms and fine filaments provide recognizable structures for such questions. Researchers can observe what the animal does with them and then investigate the cells and signals controlling their use. An observer can identify an action before knowing the neural circuit producing it. Detailed measurements of movement and tissue then allow the functional question to be investigated mechanistically.



Carlson and Birkett (2022), Fig. 1.7, p. 24. Related structures with different biological functions.



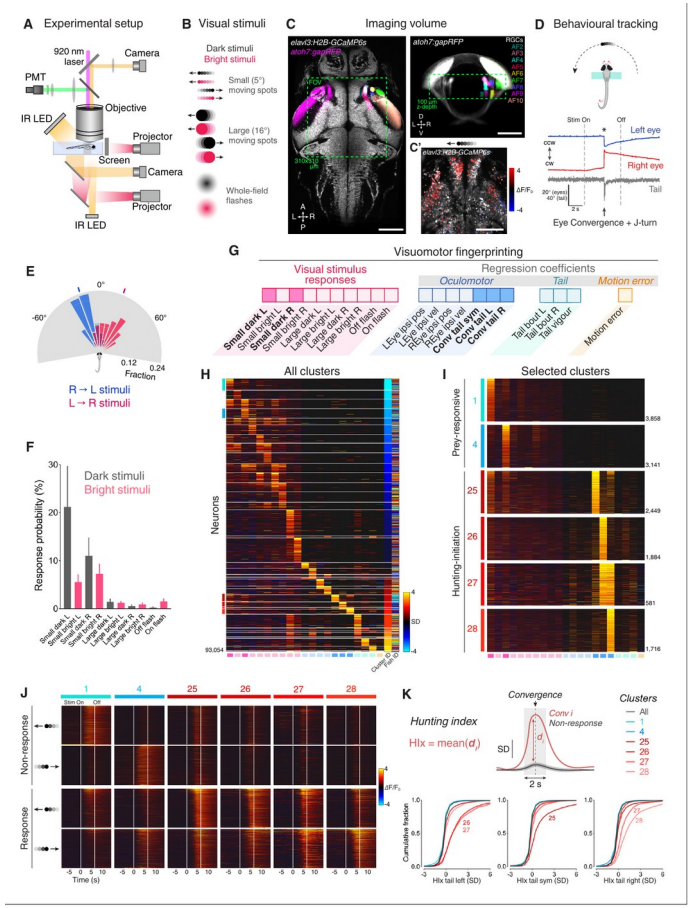
Visual context: MBARI, natural-history footage. Vampire squid (Vampyroteuthis infernalis).

Hunting begins with coordinated eye and tail movements

A hunting zebrafish turns its eyes toward a shared frontal region and moves its tail. **Eye convergence** is this inward rotation. A **saccade** is a rapid eye movement. Convergence brings the two eyes toward an overlapping frontal region. Tail movement can then orient or propel the fish while it attends to a potential prey item.

Antinucci and colleagues presented small moving spots to larval fish. About 92% of convergent saccades occurred with tail movement. The average eye-to-tail delay was about 12 milliseconds. The short delay places the eye and tail events within one rapidly coordinated response. Recording both distinguishes the start of visual orienting from the following movement of the body.

Calcium imaging measures activity-related fluorescence inside cells. Recording it alongside eye and tail movements identifies which neurons become active as the fish begins the hunting sequence. An indicator converts changes inside cells into a measurable light signal. Comparing that signal with movement times helps identify neurons active near the beginning of the behavioral sequence.



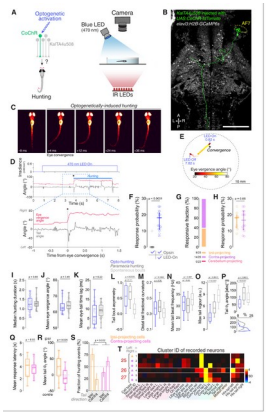
Antinucci et al. (2019), Fig. 1(A–K). Hunting-related calcium activity, eye convergence and brain anatomy.

Activating one pretectal cell can recruit hunting

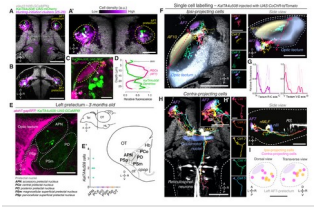
The **pretectum** is a visual-processing region near the midbrain. Antinucci and colleagues placed **CoChR**, a light-sensitive channel, in individual pretectal neurons. The researchers could locate the selected cell and stimulate it in an otherwise functioning animal. This made its contribution testable without activating every neuron in the brain.

Blue light activated the selected cell. Some fish converged their eyes and produced directed swimming bouts even without prey. A single-cell intervention could start several parts of the hunting sequence. The response included several coordinated movements rather than only a local muscle contraction.

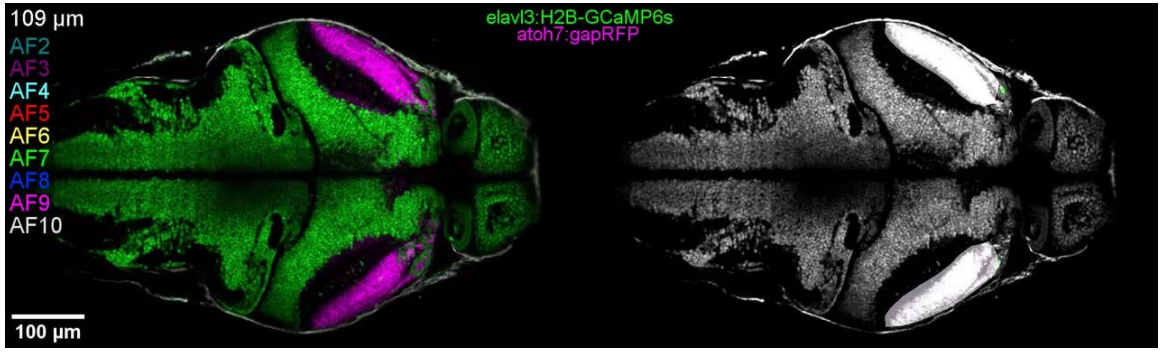
The anatomy recording traverses the fluorescently labeled brain and retinal projection fields. These are places where visual inputs enter brain tissue. Identifying them helps locate the cells whose activity can recruit hunting. Retinal neurons carry signals from the eye into the brain.



Antinucci et al. (2019), Fig. 5(A–T). Single-neuron optical activation and hunting-like kinematics.



Antinucci et al. (2019), Fig. 3(A–I). Single pretectal-cell morphology and projection classes.



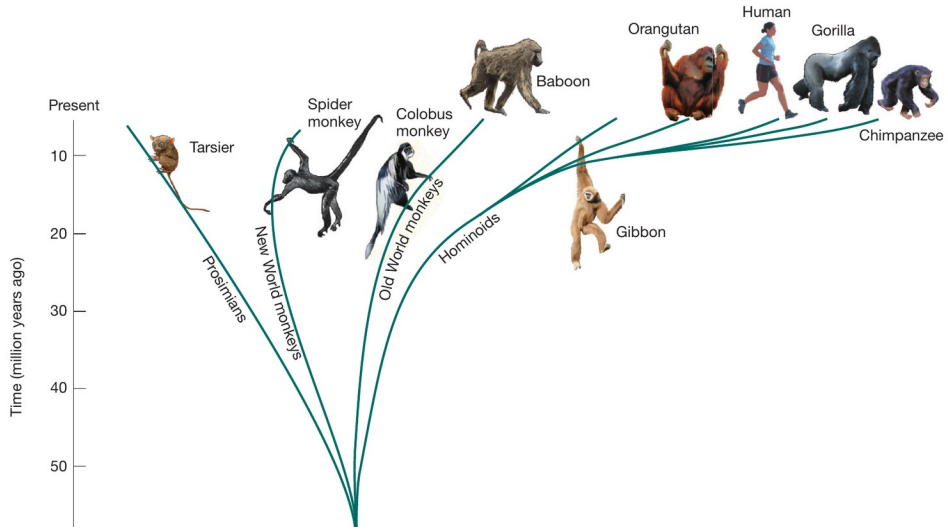
Anatomy context: Antinucci et al. (2019), Video 1. Zebrafish brain and retinal projection fields.

Mutation supplies new inherited variation

A **mutation** is a change in genetic information. Mutations can alter proteins, development, or the regulation of cellular processes. They provide one source of inherited variation. A change in a protein can alter what that protein does in a cell. A regulatory change can alter where, when, or how much of it is produced. The resulting cellular change can affect how the nervous system develops or how its cells respond to incoming signals.

A mutation can have little effect, reduce function, or change a trait in a way that matters under particular conditions. Its consequences depend on what changes and where the gene is used. A variant affecting one cellular process may influence several aspects of development. The same change can have different behavioral consequences depending on the systems that use the affected process.

Natural selection changes the frequency of variants through differences in survival and reproduction. Mutation produces variation; selection changes its distribution across generations. These are different processes. Selection does not create the mutation in response to a need. It changes how often an existing heritable variant occurs when that variant affects the number of descendants.



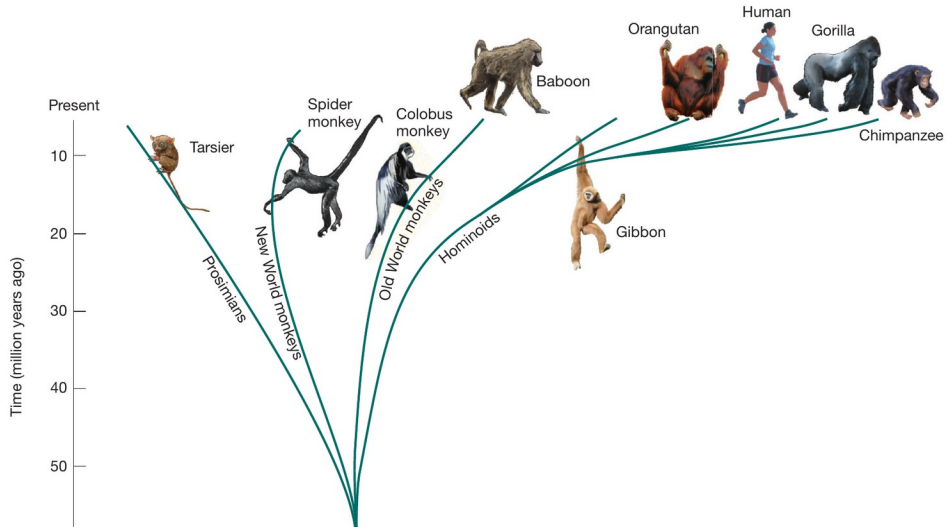
Carlson and Birkett (2022), Fig. 1.8, p. 26. Evolutionary relationships among primates.

Primate brains share an evolutionary ancestry

Primates include monkeys, apes, and humans. Related species share inherited traits, while their anatomy and behavior also differ. Shared ancestry explains why some structures recur across species. An anatomical feature shared across species may have been inherited from their common ancestor. A difference among species can instead prompt questions about changes after their lineages separated.

A **phylogeny** represents evolutionary ancestry as branches. Branch points mark the divergence of ancestral populations. Humans occupy one branch among other living primates. The branches describe descent rather than a ladder toward humans. Each living species has its own history of changes since the shared ancestral population divided.

Comparisons can separate a broadly shared ability from a specialization. Researchers can ask which vocal, motor, or learning abilities occur in several primates and which differ among their lineages. A comparison should define the behavior in the same measurable way. Similar sounds or movements can then be examined for differences in perception, learning, and neural organization.



Carlson and Birkett (2022), Fig. 1.8, p. 26. Evolutionary relationships among primates.

Brain mass and neuron number are different measurements

Brain mass measures how much tissue an animal has. Neuron number measures how many nerve cells it contains. A larger brain does not necessarily contain a proportionally larger number of cortical neurons. Brain tissue also contains supporting cells, blood vessels, and neuronal processes. Measuring its total mass combines these components, whereas counting neurons answers a more specific anatomical question.

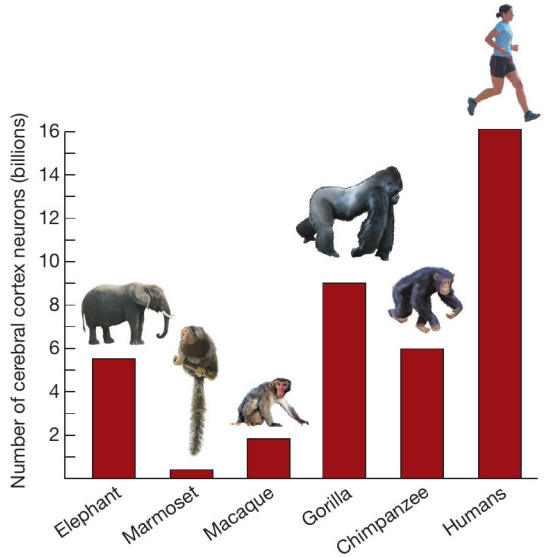
Species differ in neuron size, spacing, and the proportions of different tissues. These differences change how many cells fit within a given brain mass. Two equal masses can therefore contain different numbers of cells. Comparing cell density and tissue composition helps explain why size alone does not specify the available neuronal population.

The mammalian comparison therefore uses separate measurements of size and cell number. Explaining a behavioral ability also requires examining what those cells do and how they communicate. Cells must receive information, process it, and send outputs through appropriate pathways. Their number is an anatomical measurement, while their activity and wiring determine how a task is performed.

Figure 1.10 Comparison of Mammalian Brains

Species with more complex behaviors have brains with more neurons that are available for behavior, learning, remembering, reasoning, and making plans. Primate brains contain more neurons per gram than rodent brains and more neurons in the cortex.

Source: Herculano-Houzel, S., and Marino, L. (1998.) A Comparison of Encephalization between Odontocete Cetaceans and Anthropoid Primates. *Brain, Behavior and Evolution, 51*(4), 230–238.



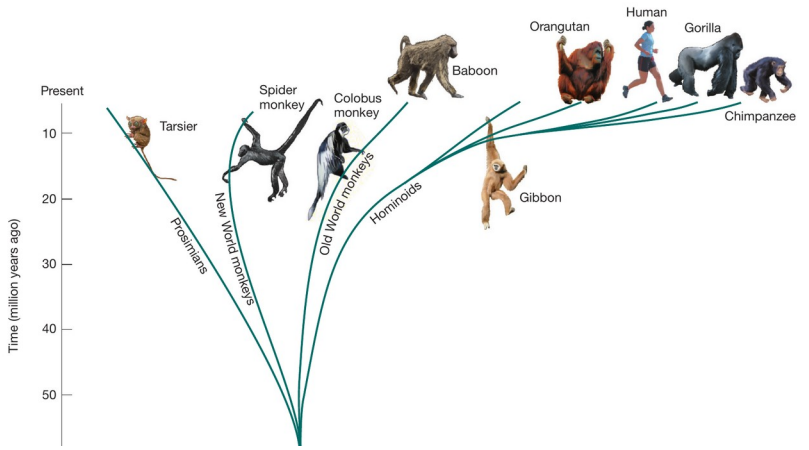
Carlson and Birkett (2022), Fig. 1.10, p. 27. Cortical neuron numbers across mammals.

Extended development contributes to human brain growth

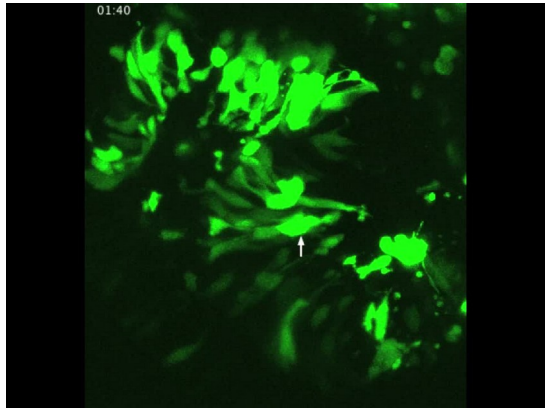
Human brain development continues well after birth. Neurons grow processes, synapses form and change, and neural pathways mature. Experience influences which patterns of activity recur. A pathway becomes useful as its cells communicate in organized patterns. Growth alone is therefore different from acquiring the ability to perform a coordinated perceptual or motor task.

Developing tissue also contains **precursor cells**, cells that can divide and produce more specialized cells. Their growth and differentiation contribute to the developing nervous system. Some precursor cells continue dividing while others begin specializing. The resulting mixture of growing and differentiating cells changes as the tissue progresses through development.

Lancaster and colleagues recorded nuclei moving within a **cerebral organoid**, a three-dimensional culture reproducing selected features of early brain development. The arrow follows a nucleus through tissue, making one developmental process visible. The publisher's timestamps track developmental movement over hours.



Carlson and Birkett (2022), Fig. 1.8, p. 26. Evolutionary relationships among primates.



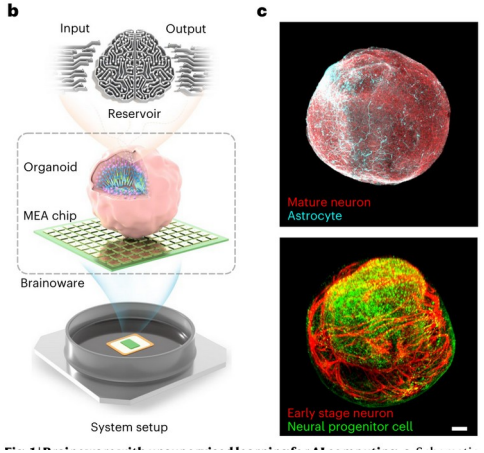
Development context: Lancaster et al. (2013), Supplementary Video 1. Nuclear movement in cerebral organoids.

An organoid transforms a changing electrical input

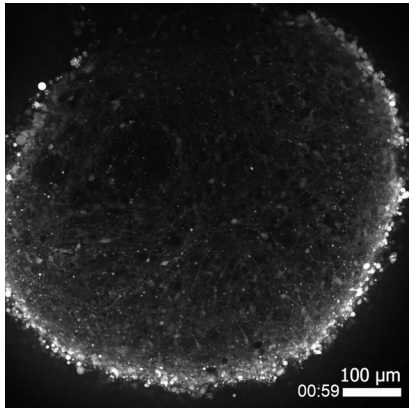
Cai and colleagues placed a human brain organoid on an electrode array. A **brain organoid** is a three-dimensional culture containing cells with selected features of developing brain tissue. The tissue contained neurons and other cell types within a three-dimensional structure. The array delivered inputs at selected contacts and sampled the electrical responses that followed.

Electrical inputs produced changing patterns of neural activity. This tissue served as a **reservoir**, a system that transforms an input into a pattern a separate decoder can use. Different inputs can evoke different patterns across the tissue and over time. These changing responses provide features that a decoder can use to distinguish one supplied input from another.

An electronic **readout** converted recorded responses into an answer. The tissue produced the response pattern; the trained decoder assigned that pattern to a category or prediction. The readout performs the task-specific assignment. A category answer therefore comes from using the tissue's response as data for the trained electronic decoder.



Cai et al. (2023), Fig. 1B–C. Organoid-electrode arrangement and neuronal tissue markers.



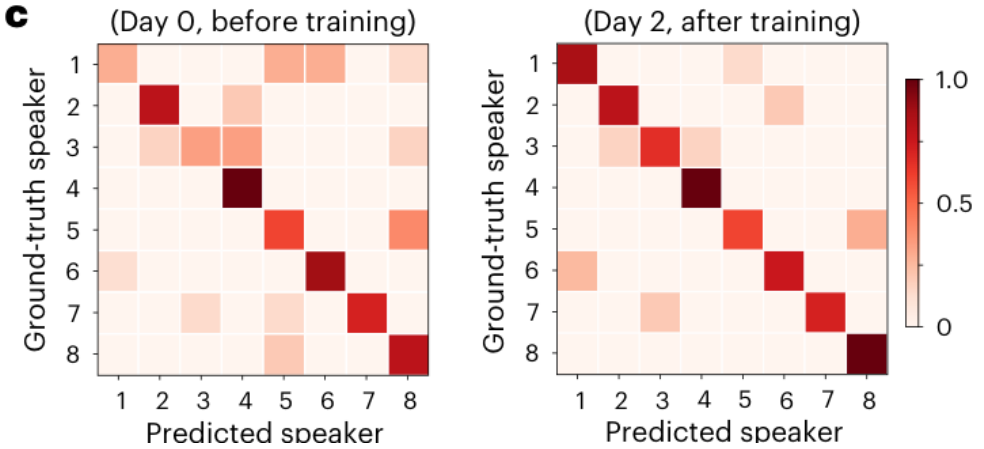
Cai et al. (2023), Supplementary Video 2. Confocal calcium activity (publisher 10× playback).

Speech classification uses a trained readout

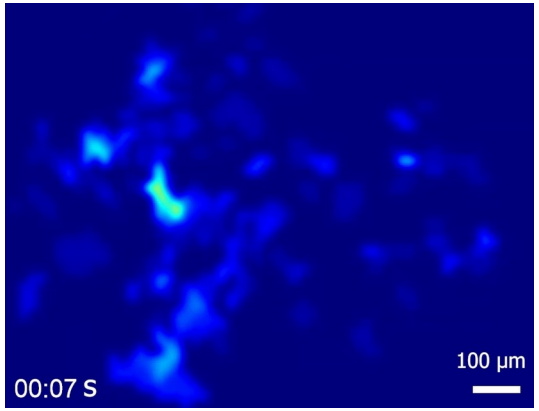
Brainoware classified recordings from eight speakers. Japanese /a/ and /e/ vowel sounds were converted into electrical stimulation patterns, which entered the organoid through electrodes. The sound was encoded rather than played to ears. Features of the recording became patterns delivered through the electrode array, giving the tissue an electrical representation of the input.

A computer decoder learned to assign the recorded tissue responses to speaker labels. The task combined repeated tissue stimulation with training of this electronic readout. The assigned labels identified the speaker. The decoder used response features to distinguish those labels, while repeated exposure supplied the organoid with the encoded recordings during training.

Accuracy increased from about 51% to 78% after four **epochs**, passes through the training data, over two days. The result measured speaker classification by the tissue-and-decoder system, not spoken language produced by the organoid. Accuracy counts the proportion of speaker labels assigned correctly.



Cai et al. (2023), Fig. 3C. Speaker classification before and after training.



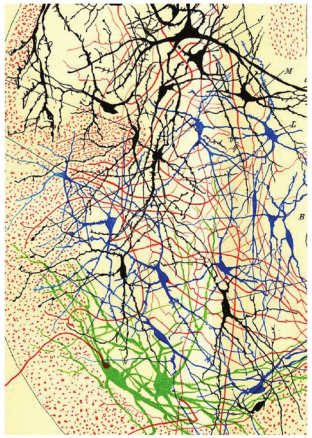
Cai et al. (2023), Supplementary Video 1. Organoid spike-activity heat map (publisher 4× playback).

Animal research includes care and ethical oversight

Animal research requires appropriate housing, veterinary care, and ethical oversight. A proposed experiment must justify its scientific purpose and describe how the animals will be cared for. Care requirements depend on the species and the procedure. Monitoring feeding, health, and participation helps identify changes in the animal's condition throughout an experiment.

The **three Rs** are replacement, reduction, and refinement. Replacement uses suitable alternatives; reduction avoids unnecessary animal use; refinement improves procedures and care to reduce harm. The choice of a method affects how these principles can be applied. A refined procedure can preserve the research question while changing handling, housing, or the demands of the task.

The sea-lion footage provides an example of a trained movement in managed care. Researchers also measure feeding and voluntary participation when evaluating cognitive activities. Welfare can be examined through repeated observations of the animal's behavior. A trained action can be measured alongside daily behavior outside the task.



Carlson and Birkett (2022), Fig. 1.5, p. 21. Cajal's drawing of Golgi-stained neurons.



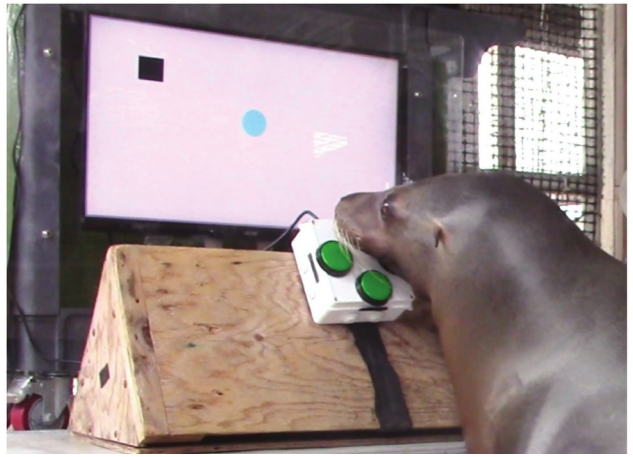
Visual context: Blackpool Zoo (2009). Bimbo balancing a ball.

Sea lions learn to control a computer cursor

Winship and colleagues trained sea lions to operate EVE, an underwater computer interface. A separate display presented a cursor and targets. Directional buttons controlled cursor movement. The controller registered contact away from the displayed target. The animal therefore controlled movement through a device rather than moving its muzzle directly to the target location.

The sea lion contacted buttons with its muzzle. Each direction moved the cursor across the screen. Reaching a target produced a success signal and reinforcement. Each contact had a direction and a duration. Selecting a button started movement, while releasing or changing it altered the cursor’s progress toward the visual goal.

Training established the button actions before presenting more complex tasks. The animal learned to watch the cursor, choose an input, and stop or change that input as the cursor approached the target. Successful contact made the consequence available immediately after the relevant action. The animal could then practice the same input rules as the task’s visual requirements became more complex.



Winship et al. (2023), Fig. 7. Sea lion operating the cursor task



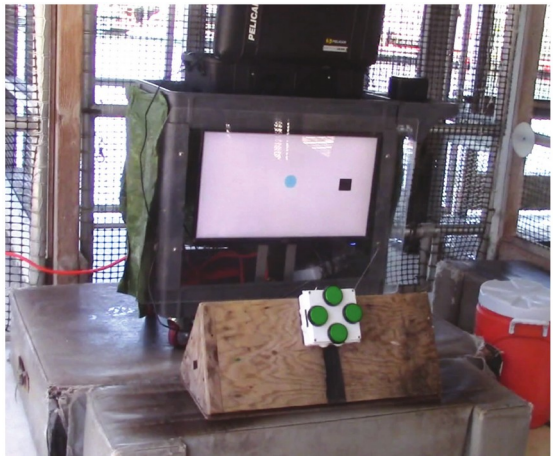
Winship et al. (2023), EVE excerpt 0:00–0:13. Brief button contact.

A display and a touchscreen are different interfaces

A **touchscreen** records contact on the display itself. EVE used a separate monitor and directional controller. The sea lion moved a cursor rather than touching the target's displayed location. Touching a location and controlling movement toward it are different actions. A separated display and controller require the animal to use an input rule rather than reach directly to the goal.

A button press changed the cursor's direction. Contact duration affected how far it moved. The animal had to monitor movement elsewhere while operating the controller. A longer contact can keep the cursor traveling after a brief contact would stop. The animal must use its view of the changing screen to decide when to release the input.

The input and the visual goal occupied different places. Successful performance required selecting and timing button contacts so the cursor arrived at the target. Progress on the display supplies feedback about the current action. Selecting another direction can correct the route before the cursor reaches the intended destination.



Winship et al. (2023), Fig. 1. Separate monitor and directional controller



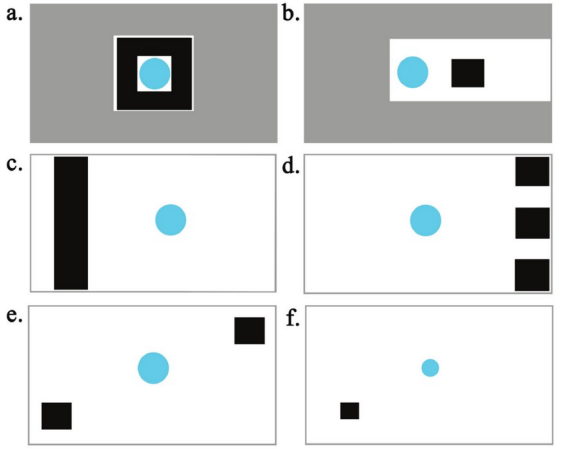
Meet_EVE_HD, 70–84 s. Sample and choice display.

Cursor contact determines the trial consequence

A **contingency** specifies which event produces a consequence. In EVE's cursor task, touching the target with the cursor produced a success tone and reinforcement. The consequence depends on the software detecting cursor contact, not merely any controller touch. This defines an observable event that can be scored in the same way across trials.

A blue circle served as the cursor, with black rectangular targets on a white background. Target location and cursor size changed during training. A smaller cursor or a different target position changes how precisely movement must be controlled. The animal still uses the same buttons, but must adjust the timing or direction of its inputs.

Software recorded button directions, contact durations, and response times. The researchers could measure both success and the sequence of actions that moved the cursor into the target. An input log identifies when the animal pressed and released each control. It can therefore describe the actions producing a success rather than record only the final outcome.



Winship et al. (2023), Fig. 2. Successive cursor-training task displays



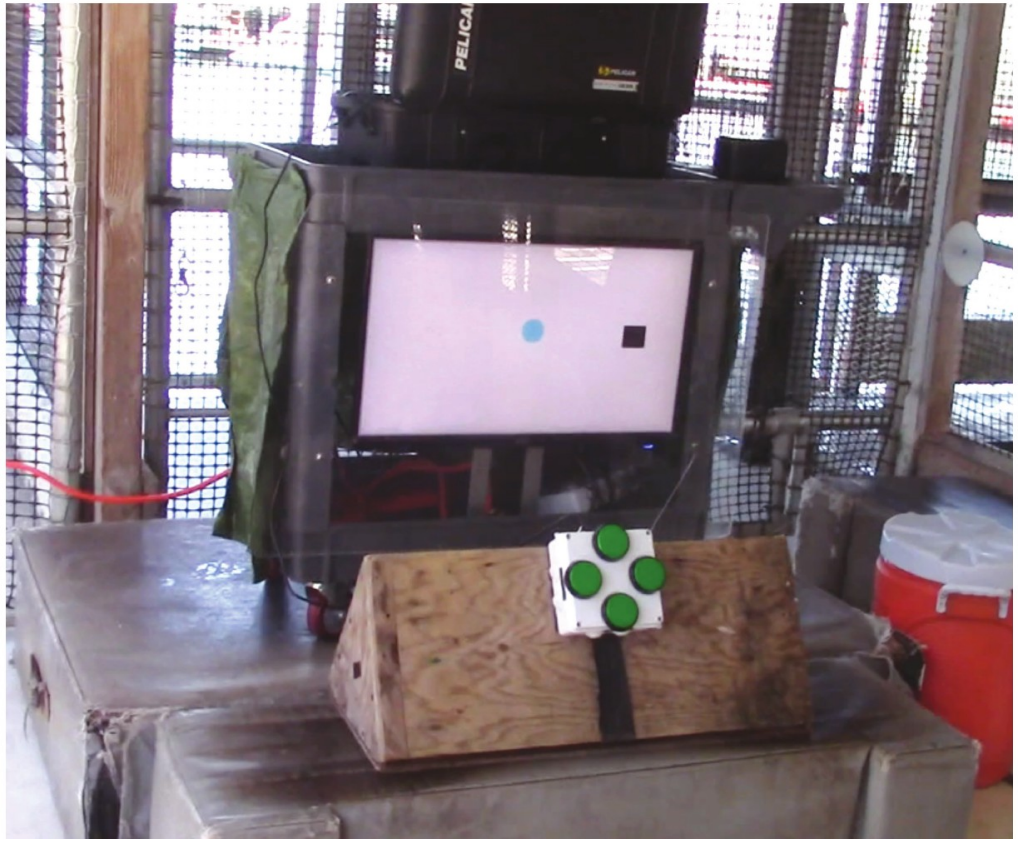
Meet_EVE_HD, 84–101 s. Response and trainer reward.

A success tone becomes a conditioned reinforcer

A **conditioned reinforcer** is a signal that gains reinforcing value through pairing with other consequences. During EVE training, a success tone was paired with food and other reinforcements. The tone initially occurs with consequences the animal already receives as reinforcement. Through repeated pairing, the sound can mark that a successful event has just occurred.

Early in training, a trainer could deliver the tone manually. Later, software generated it when the cursor touched a target. The sound marked the successful event immediately. Software delivery makes the timing depend on the game event. The tone can occur as soon as the cursor reaches the target, even when the additional reinforcer is delivered later.

Food, ice, and trainer encouragement followed successful responses. The tone identified which action had earned reinforcement, helping the animal learn the task even when the trainer did not visibly signal the correct movement. A prompt signal identifies the successful event among several recent actions.



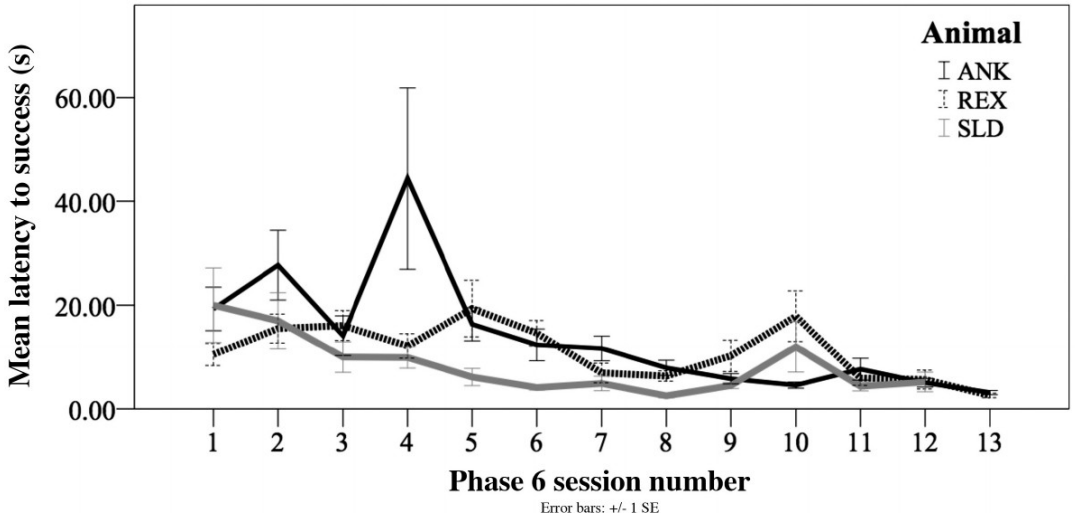
Winship et al. (2023), Fig. 1. Apparatus used for tone-conditioned gameplay

Target-contact latency declines during training

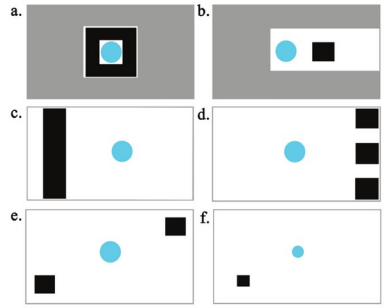
Latency is the time from a defined starting event to a response. EVE measured the interval from the first button contact to cursor contact with the target. The interval begins after the first contact, so it describes performance once the animal starts controlling the cursor. It does not include the time spent deciding whether to begin.

Across later training sessions, this interval decreased for each sea lion. Animals reached the target more quickly after starting to move the cursor. A smaller interval means the animal reaches the goal sooner after starting. Comparing the same training phase makes this improvement distinguishable from simply using an easier early task.

A faster trial can involve a straighter route, better-timed release, or less ineffective input. Recorded button sequences identify which actions occurred while the cursor traveled toward the target. Two trials can have the same final outcome but different routes and pauses. The input log describes those differences and allows the time measure to be interpreted more specifically.



Winship et al. (2023), Fig. 4. Latency across final training sessions



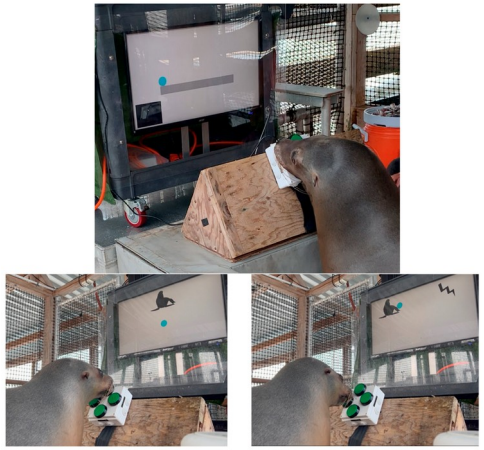
Winship et al. (2023), Fig. 2. Single-target task underlying the latency measure

Maze choices determine where the cursor falls

In EVE’s maze game, the sea lion chose left or right movements. The cursor traveled along platforms and could fall toward a target below. A horizontal movement determined where the falling cursor began its next step. A choice on one platform therefore affected which positions became reachable on the platform below.

A horizontal choice changed where the cursor dropped. A successful route placed it above the target; another route could send it away from the goal. The result was not determined by moving toward the screen’s nearest visible point alone. The animal had to select a route whose subsequent drop could place the cursor at the target.

The maze made an action’s later consequence visible. Introductory trials used simpler layouts before more complex arrangements. The animal had to choose movements according to where the cursor would go next. Simpler layouts reduce the number of possible routes. More complex layouts require using the acquired left and right controls in a sequence suited to the arrangement of platforms.



Roberts et al. (2023), Fig. 3. Maze training arrangement and matching alternatives



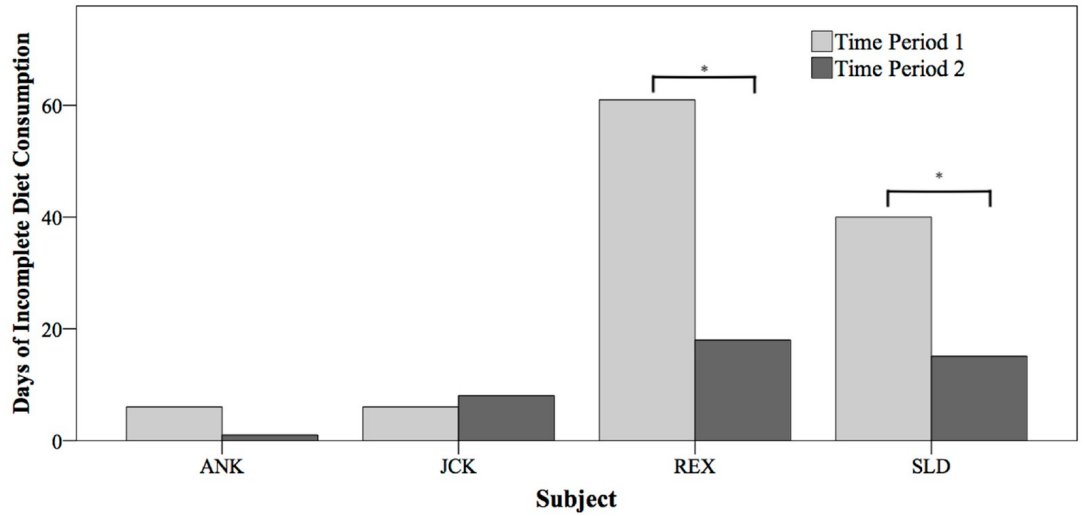
Meet_EVE_HD, 53–59 s. Maze gameplay.

Feeding records change after cognitive enrichment

Winship and colleagues compared feeding records before and after EVE was introduced. **Incomplete consumption** meant leaving some of the offered daily ration uneaten. Feeding is a daily behavior that can be recorded repeatedly. These observations provided an outcome outside the game rather than another measure of cursor speed or controller use.

Unfinished-ration days fell from 61 to 18 for REX and from 40 to 15 for SLD. Days affected by clinical illness were treated separately in the analysis. A day with part of the ration unfinished counted toward incomplete consumption. Separating clinical-illness days distinguishes those records from the feeding observations analyzed for the enrichment comparison.

Average consumption rose from about 97% to 99% for REX and 98% to 99% for SLD. The enrichment period coincided with more complete feeding. Daily records measured behavior outside the computer trials. The percentage consumed measures how much was eaten, while the day count measures how often a ration was unfinished.



Winship et al. (2024), Fig. 2. Incomplete-diet-consumption days across two periods



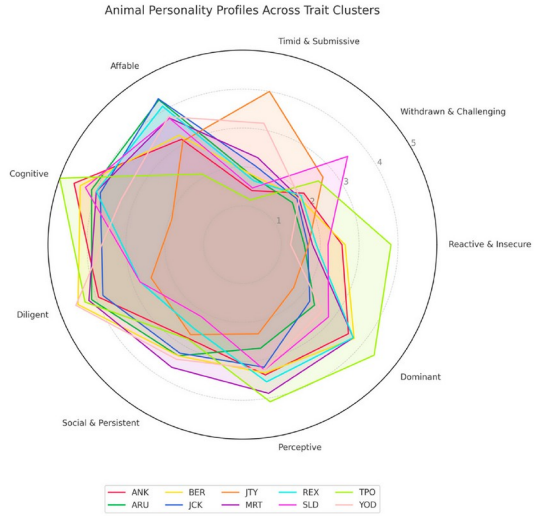
Winship et al. (2024), Fig. 1. EVE session context for the feeding records

Rated perceptiveness is associated with faster responses

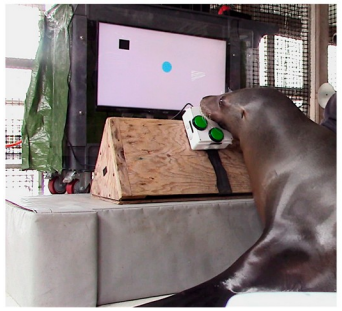
Ramos and Winship asked familiar trainers to rate the sea lions' recurring behavioral tendencies. **Personality traits** describe how an individual tends to behave across situations. The ratings summarized patterns observed outside a single trial. The computer task then provided separate records of how quickly the animal reached a target and how many inputs it used.

Higher Perceptiveness ratings accompanied faster target-contact responses in both initial and later sessions. Higher Dominance ratings accompanied faster initial responses and fewer initial button presses. The associations concern rated traits and measured performance. A consistent rating can therefore be compared with a specific action measure across more than one stage of training.

Speed and input count measure different aspects of performance. A sea lion can reach a target sooner, use fewer contacts, or change both. The study compared these recorded actions with the trainers' ratings. Counting contacts and timing the response make these differences measurable.



Ramos & Winship (2025), Fig. 5. Trait profiles related to response measures



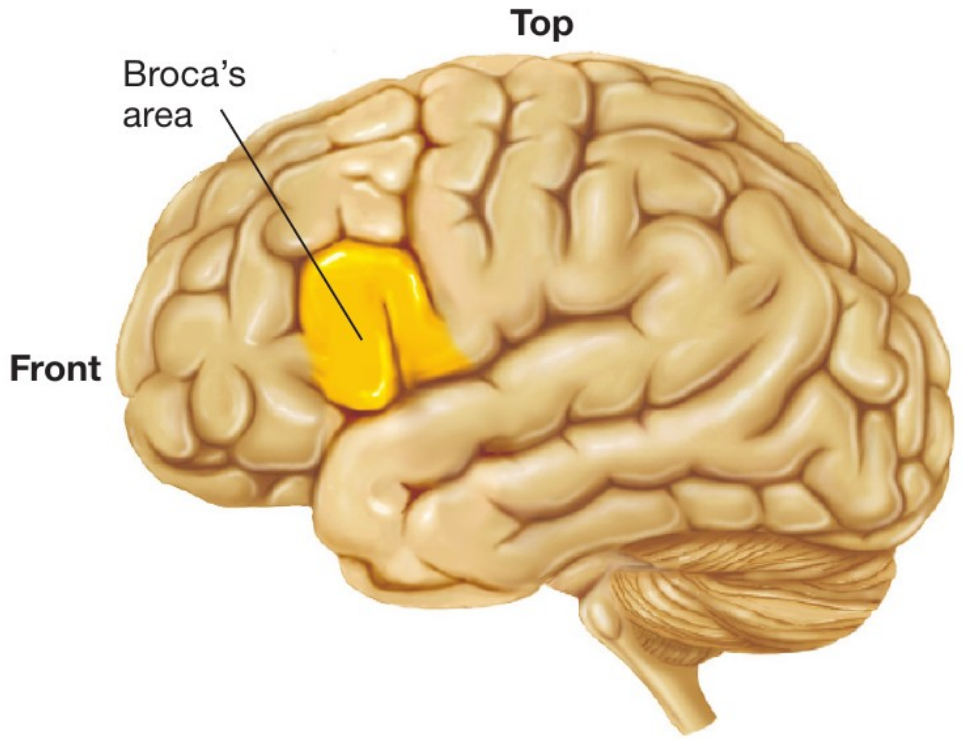
Ramos & Winship (2025), Fig. 1. Computerized task used for response measurements

Human research includes consent and privacy

Informed consent means that a person understands the proposed research and voluntarily agrees to participate. Researchers explain the procedures, likely burdens, and the participant's right to withdraw. Consent is a continuing choice rather than only an initial signature. A participant can ask questions or withdraw without being required to complete every planned procedure.

Privacy protects personal information, including clinical history and brain recordings. Researchers limit who can access identifying data and how those data are stored or shared. A recording may contain information about health as well as the task. Restricting identifying information reduces the chance that a person's clinical details are exposed unnecessarily.

Clinical research may include people with neurological injuries or conditions. Their circumstances affect how procedures are explained and what participation requires. Ethical review evaluates these demands before the study begins. The ability to understand information or communicate a decision can vary with the condition.



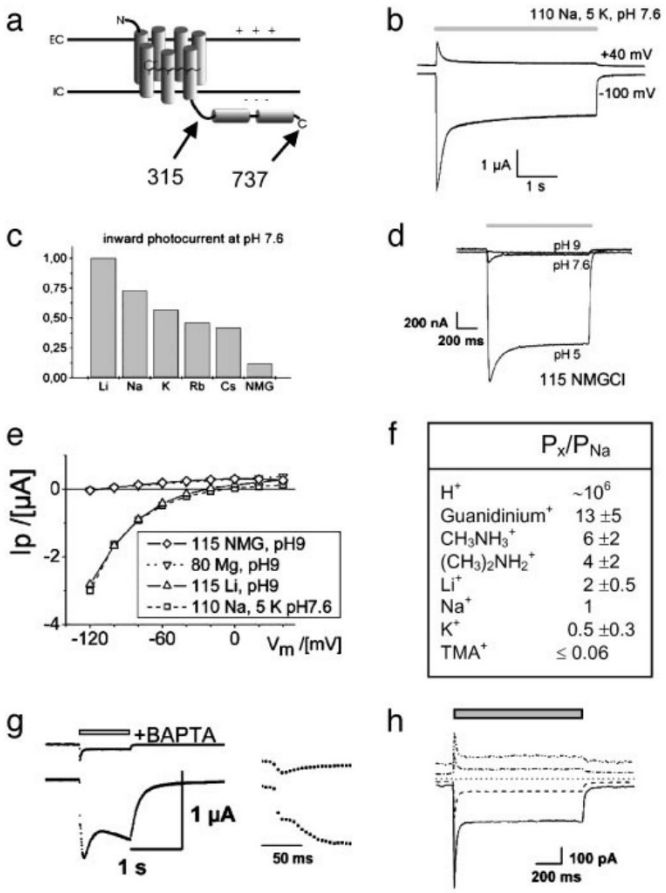
Carlson and Birkett (2022), Fig. 1.4, p. 21. Broca's area in the left hemisphere.

Optogenetics makes selected neurons responsive to light

Optogenetics places a light-sensitive protein in selected cells. An **ion channel** is a protein pathway through the membrane. Opening the channel allows charged particles called ions to cross. Targeting determines which cells receive the protein. Illumination determines when those cells experience the optical input, separating cell selection from the timing of the experimental change.

Channelrhodopsin-2, or ChR2, comes from an alga. Its light-absorbing molecule, **retinal**, responds to blue light. The protein opens a channel through which positive ions can move. Opening the channel provides a path across a membrane that normally restricts ion movement. The resulting current changes the cell's voltage and can alter its tendency to fire.

Light can then change the firing of selected neurons at a chosen time. Karl Deisseroth, Peter Hegemann, and Georg Nagel received the 2026 Nobel Prize in Physiology or Medicine for optogenetics. A behavioral experiment can compare the animal's actions during and between illumination periods. This makes the manipulation useful for testing what the selected cells contribute to an action.



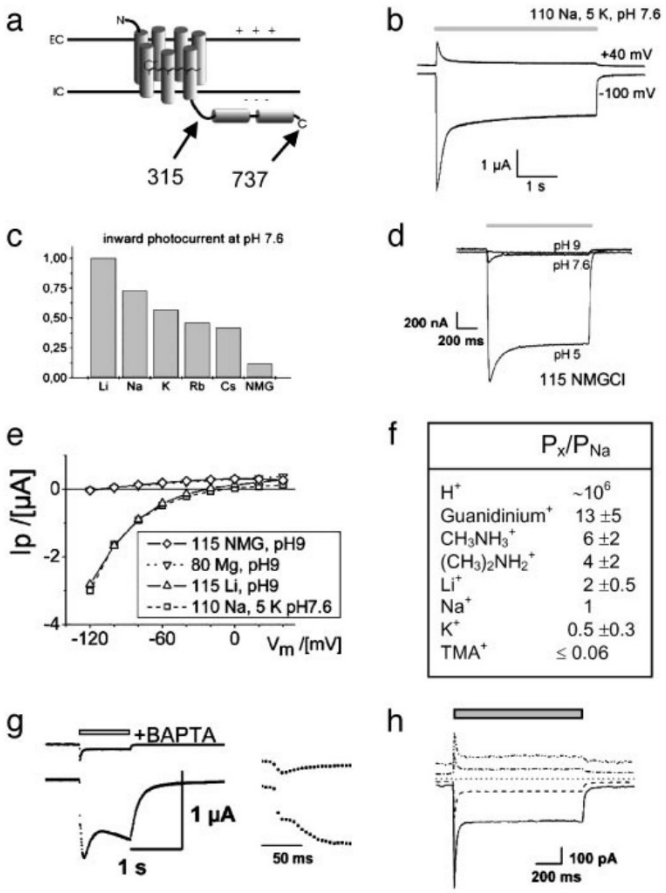
Nagel et al. (2003), Fig. 1(a–h). Original ChR2 currents under ion substitution and voltage clamp.

Ion movement changes a cell’s electrical state

An **ion** carries electrical charge. Sodium, potassium, and calcium ions are positive; chloride is negative. Different concentrations occur inside and outside a neuron. The membrane separates solutions with different ion concentrations. An ion’s charge also determines how the voltage across that membrane influences its movement through an open channel.

An open channel allows ions to cross the membrane. Concentration differences and electrical forces determine their movement. Moving charge changes the **membrane potential**, the voltage difference across the membrane. The direction of movement depends on both forces together. Opening a channel does not always produce the same current because the cell’s voltage and ionic conditions can differ.

A channel’s **selectivity** determines which ions can pass. Changing the available ions changes the measured current. In optogenetics, light opens the channel, ions move, and the neuron’s voltage changes. A sodium-conducting pathway and a chloride-conducting pathway can have different effects. Identifying the passing ions helps explain why different light-sensitive proteins change neuronal activity differently.



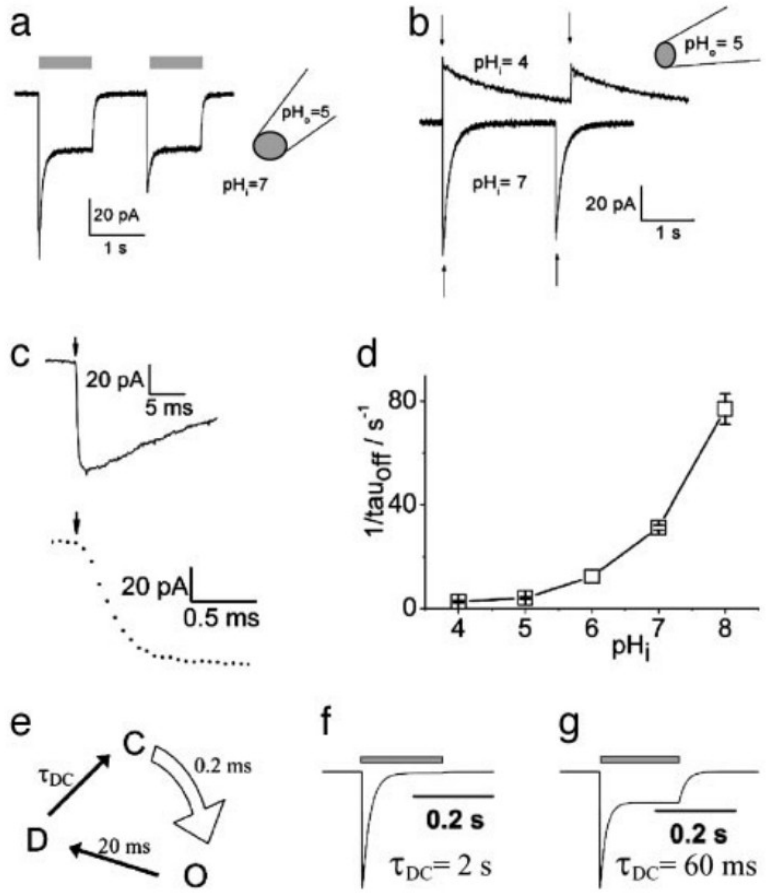
Nagel et al. (2003), Fig. 1(a–h). Currents and selectivity under controlled salt solutions.

Channel opening gives a neural signal its time course

Gating is the opening or closing of an ion channel. Light-gated channels respond to illumination. Voltage-gated channels respond to the membrane voltage. The activating signal and the resulting current are different events. The protein must change state before ions pass, and its return to a closed state stops that pathway.

A flash can open a channel rapidly, while closing takes longer. The electrical current can therefore continue after the brief flash ends. Opening and closing determine the input's time course. A fast-opening channel can respond soon after the flash. A slow-closing channel can keep influencing the cell after illumination ends, extending the input beyond the stimulus.

A neuron receives many inputs at different times. A longer-lasting current overlaps more of those inputs. Channel timing affects whether their combined effect brings the neuron to firing threshold. Inputs that arrive close together can combine before the membrane returns toward its earlier voltage. Their timing therefore affects whether the cell reaches the level needed to fire.



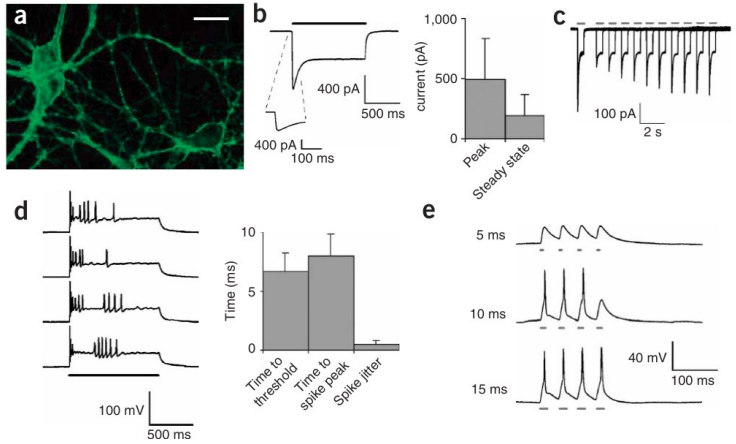
Nagel et al. (2003), Fig. 3(a–g). Excised-patch responses to light and intracellular pH.

Depolarization can trigger an action potential

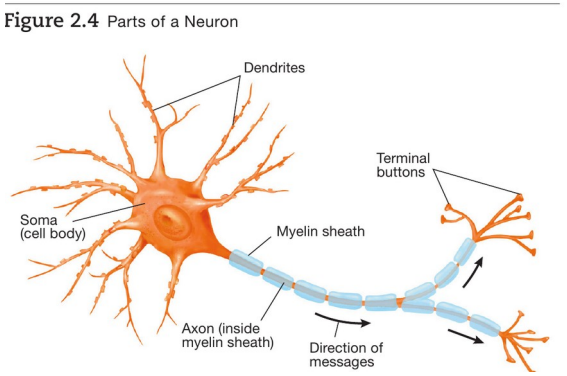
A resting neuron is usually more negative inside than outside. **Depolarization** makes this difference less negative. Inward movement of positive charge can bring the membrane to firing threshold. The amount of depolarization depends on the current and the other inputs arriving at that moment. A small input may change voltage without triggering a spike.

An **action potential** is a brief electrical signal that renews itself as it travels along an axon. Voltage-gated channels open in sequence as adjacent membrane regions change voltage. The traveling signal repeatedly activates the next region of axonal membrane. This renewal allows a short event at one location to affect terminals some distance away.

At a **synapse**, one neuron influences another cell. Many synapses release chemical messengers called **neurotransmitters**. Light-triggered firing can therefore change the activity of cells receiving the stimulated neuron's outputs. The transmitter's effect depends on the receiving receptors and cell.



Boyden et al. (2005), Fig. 1(a–e). Membrane expression, inward current and light-evoked spikes.



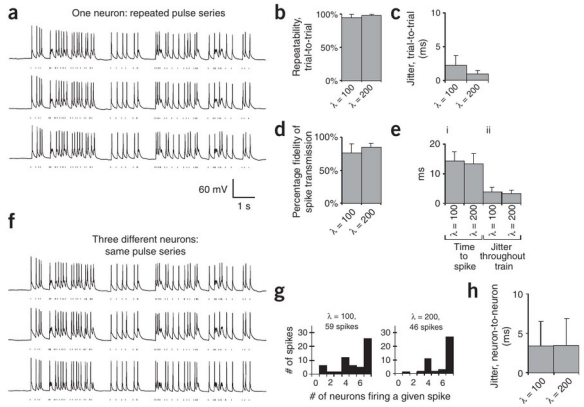
Carlson and Birkett (2022), Fig. 2.4, p. 40. Labeled cell body, dendrites, axon, and terminals.

Brief light pulses can make neurons fire

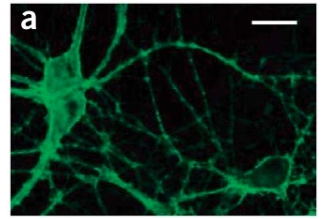
Boyden and colleagues placed ChR2 in rat neurons. Blue light opened the channel, positive charge entered, and the membrane depolarized. A short pulse could trigger an action potential. The introduced channel supplied an ion pathway controlled by illumination. Recording the resulting voltage changes allowed the researchers to measure the cell's electrical response directly.

Repeating light pulses delivered a timed series of inputs. Electrical recordings measured whether the neuron fired after each pulse. The timing of stimulation determined when it could influence other cells. The cell must respond quickly enough for separate pulses to produce separate spikes.

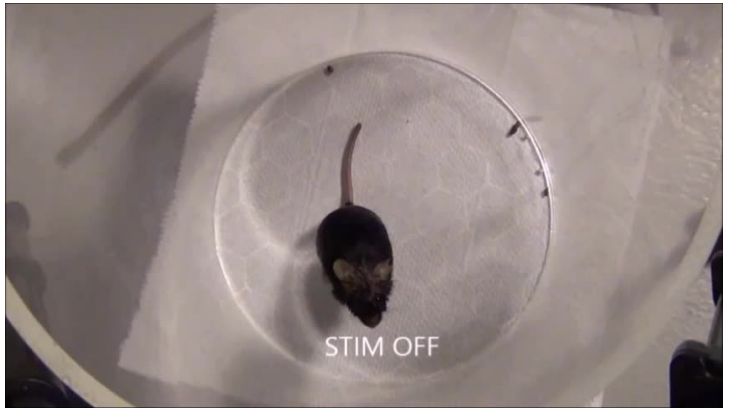
Montgomery and colleagues delivered light wirelessly to a mouse premotor region. Circling rose from about 0.4 to 2.5 turns per minute. Activating selected neurons changed the animal's direction of movement. The premotor region participates in organizing movement. Increasing its activity changed the trajectory of the whole animal, extending the manipulation from a cellular signal to a measurable action.



Boyden et al. (2005), Fig. 2(a–h). Repeated pulse trains and timing of evoked spikes.



Boyden et al. (2005), Fig. 1(a). ChR2-YFP membrane expression in a cultured hippocampal neuron.



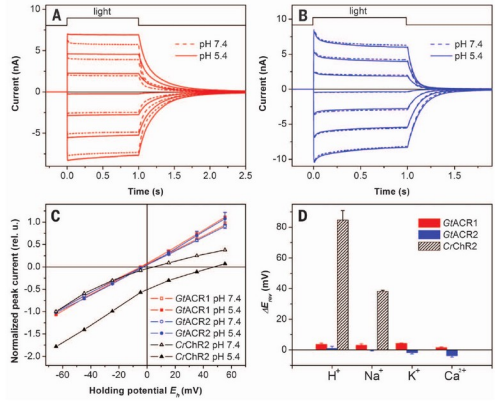
Montgomery et al. (2015), Supplementary Video 1. Wireless light delivery to premotor cortex and circling behavior.

Excitation and inhibition shape circuit activity

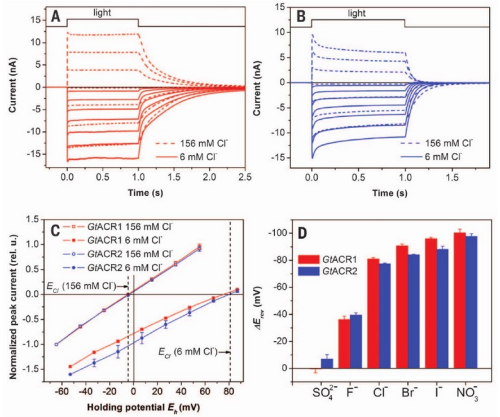
Excitation increases a neuron's tendency to fire. **Inhibition** reduces it. The neuron's output depends on the combined effects of its incoming signals. Several inputs may arrive while the membrane is already responding to earlier signals. The combined current determines whether voltage approaches or moves away from firing threshold.

The effect of a channel depends on which ions move through it. ChR2 carries positive ions. **Anion channelrhodopsins** carry negative ions such as chloride. An ion's effect depends on its concentration and the membrane voltage as well as its charge. Channel identity alone is not a complete description of the electrical response.

Optical activation and inhibition let researchers change firing in different directions. They can measure how increasing or decreasing activity changes movement, sensation, or another behavior. An experiment can increase one selected input while leaving other inputs present.



Govorunova et al. (2015), Fig. 2(A–D). Cation substitution and photocurrent reversal in GtACRs.



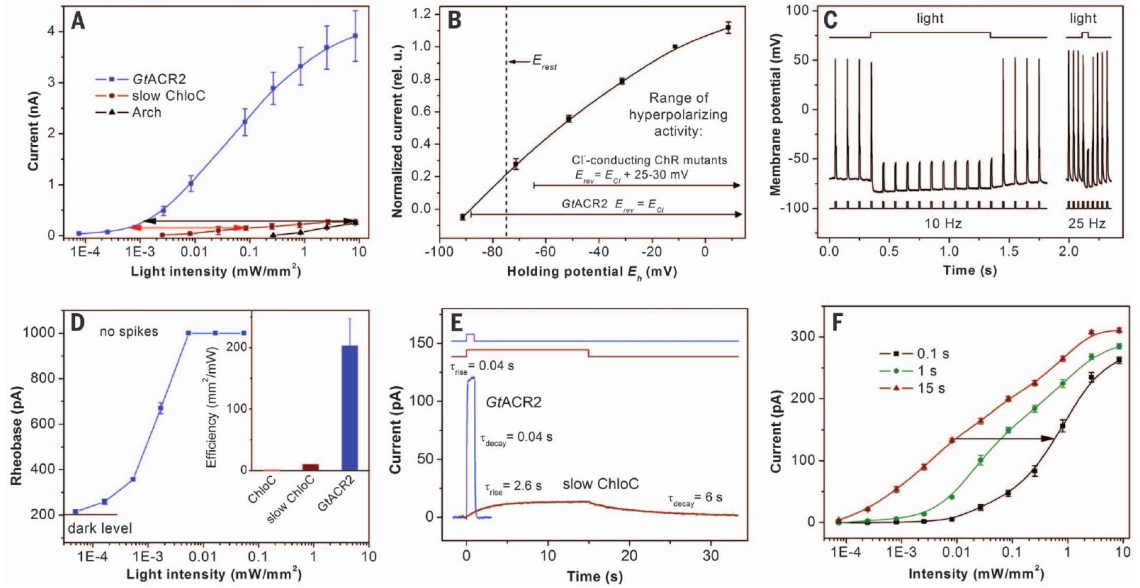
Govorunova et al. (2015), Fig. 3(A–D). Chloride substitution identifies anion selectivity.

Inhibitory signals change the probability of firing

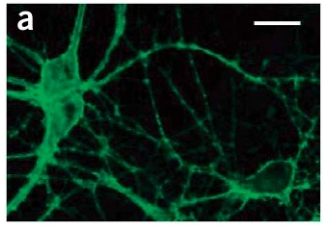
Inhibition can move a neuron's voltage away from firing threshold. It can also reduce the effect of an excitatory input by opening another pathway for charge to cross the membrane. Opening an additional ion pathway changes how incoming charge affects voltage. An excitatory signal can then produce a smaller voltage change than it would without that pathway.

Chloride movement depends on its concentration and the membrane voltage. Opening a chloride channel drives the voltage toward the **chloride equilibrium potential**, where the forces on chloride balance. At equilibrium, concentration and electrical forces balance for chloride. Away from that voltage, chloride current drives the membrane toward the value determined by those forces.

Govorunova and colleagues used a light-sensitive chloride channel to suppress firing. Spikes stopped during illumination and resumed afterward. Light controlled the periods in which the neuron sent fewer electrical signals. The resulting reduction in firing changes the signals reaching downstream cells.



Govorunova et al. (2015), Fig. 4(A–F). GtACR2 current reversal, spike suppression and light sensitivity.



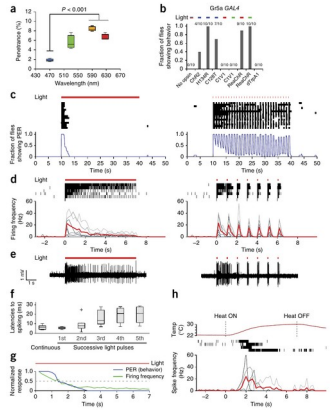
Boyden et al. (2005), Fig. 1(a). ChR2-YFP membrane expression in a cultured hippocampal neuron.

Activating fly neurons recruits specific actions

Inagaki and colleagues used **ReaChR**, a channel activated by longer-wavelength light than ChR2. They placed it in selected fly neurons, allowing red light to change those cells' activity. The protein gave targeted cells an optical input while the animal remained intact. Light could influence those neurons without physically moving the fly's feeding structure.

Activating sugar-sensing **Gr5a neurons** triggered extension of the **proboscis**, the fly's feeding structure. The mouthpart moved outward during illumination. The response began through the activated cells rather than by pushing the mouthpart. Muscles then produced the visible extension, making the action an output of the stimulated neural pathway.

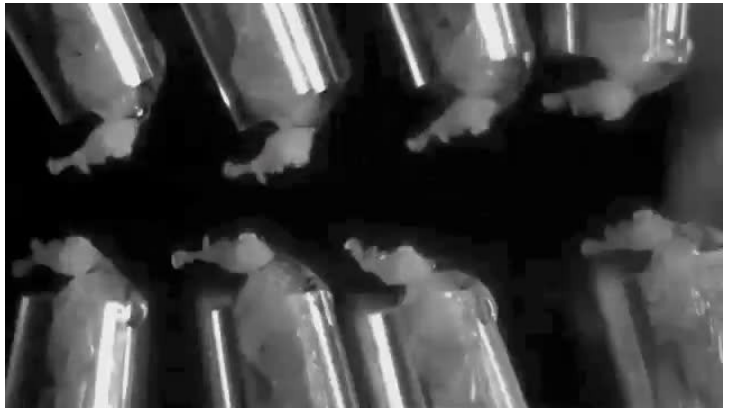
Other targeted cells produced other actions. Selecting the neuron group determines which response light can recruit. The feeding example makes the sequence concrete: activate sensory neurons, then measure the mouthpart movement. A feeding movement and a courtship movement require different muscle patterns.



Inagaki et al. (2014), Fig. 1(a–h). Opsin comparisons, cuticle transmission and gustatory-neuron responses.



Grover et al. (2020), Fig. 1(c). Head imaging window and retroreflective beads on an adult fly.



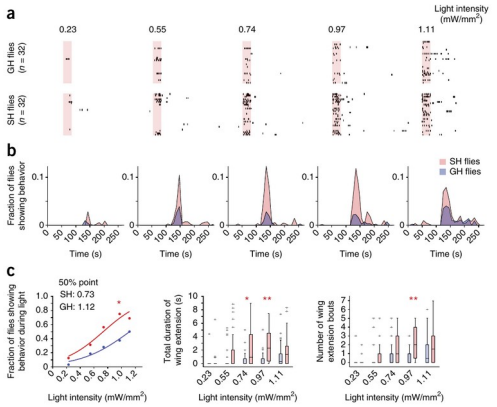
Inagaki et al. (2014), Supplementary Video 1. Light-triggered proboscis extension.

Experience changes how a fly circuit influences courtship

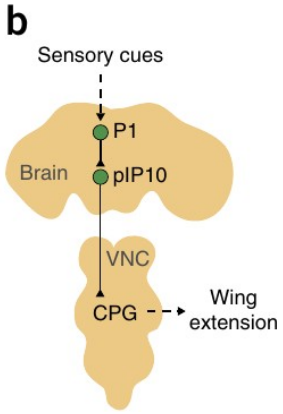
P1 neurons participate in male fly courtship. Inagaki and colleagues activated them with light and measured courtship actions, including wing extension. Courtship includes directed movements of the wings and body. The researchers measured these actions rather than treating every increase in movement as the same behavioral response.

The response differed with the fly's previous social experience. The same targeted activation could produce different behavior after different housing histories. Previous social conditions altered the response without changing the intended neuronal target. The result makes experience part of the explanation for what the circuit does during stimulation.

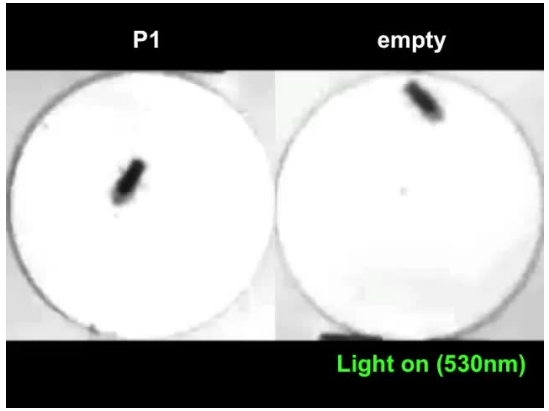
The recording also includes activation of **pIP10**, another courtship-related group. Wing and body movements identify the output. Prior experience changes the circuit's state and therefore changes the response to activation. Different neuronal groups can recruit components of a broader behavioral sequence.



Inagaki et al. (2014), Fig. 4(a–c). Social housing changes P1-evoked wing-extension sensitivity.



Inagaki et al. (2014), Fig. 3(b). Published P1 and pIP10 circuit arrangement; connectivity is proposed.



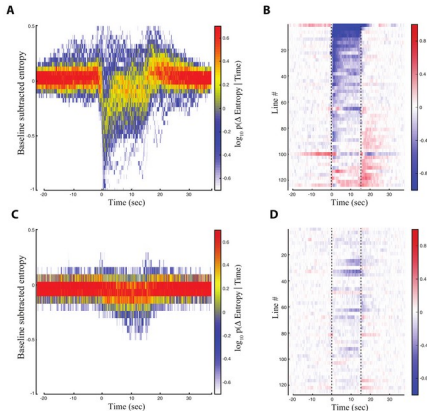
Inagaki et al. (2014), Supplementary Video 5. P1 and pIP10 activation recruits wing extension.

Fly circuits recruit recognizable movement patterns

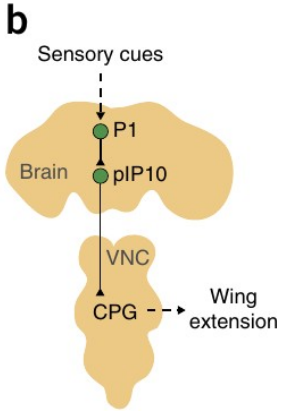
Descending neurons carry signals from the fly brain toward movement-control circuits. Cande and colleagues activated selected neuron types and measured the actions that followed. The brain's output reaches other circuits rather than directly supplying every muscle command. Those downstream circuits organize the pattern of limb and body movements.

Different targeted types favored different movement patterns. Activation did not simply increase every action equally. The limbs and body changed their organization according to the targeted cells. A behavioral pattern includes which body parts move and the order of their movements.

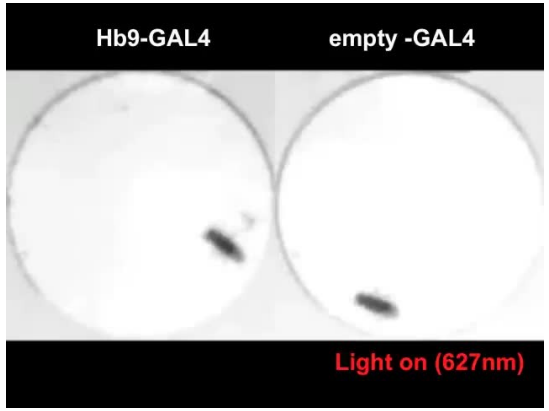
A separate Inagaki recording provides a recognizable movement example: activation with an HB9 driver produced sideways walking. The preparations differ, but both allow a specific manipulation to be compared with a visible action. Sideways walking is recognizable through the direction of body displacement and the accompanying leg movements.



Cande et al. (2018), Fig. 3(A–D). Behavioral entropy across descending-neuron lines.



Inagaki et al. (2014), Fig. 3(b). Published P1 and pIP10 circuit arrangement; connectivity is proposed.



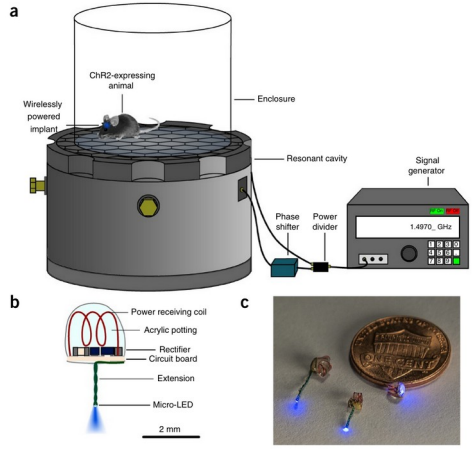
Movement context: Inagaki et al. (2014), Supplementary Video 2. Sideways walking during optical activation.

An internal implant can receive wireless optical power

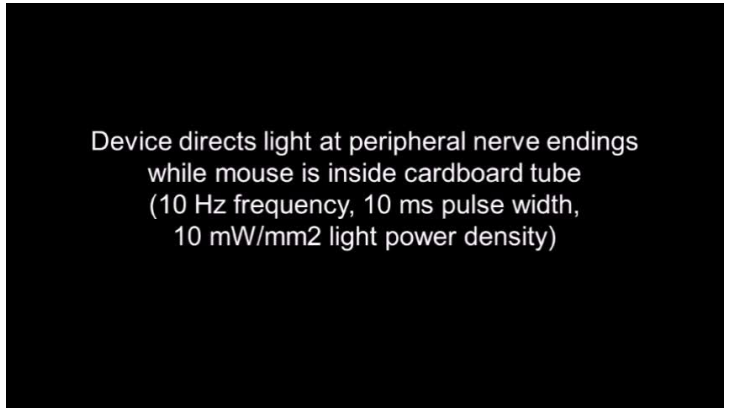
Montgomery and colleagues implanted a small light-delivery device in mice. A **micro-LED**, a miniature light-emitting diode, produced light near selected brain tissue. The emitter produced light close to the target rather than projecting it through a long external fiber. Device placement determined which tissue received the optical input.

An external radio-frequency source supplied power. A receiving coil inside the animal converted this energy into electrical power for the LED. The smallest device weighed about 20 milligrams. Energy transfer powered the emitter during the animal's movement. The receiver and LED performed different steps: one obtained power, while the other converted that power into light.

The mouse could move without an external light-delivery fiber. Running, entering a tube, and interacting with another mouse remained possible while the implant delivered optical stimulation. Removing a tether allows the animal to perform actions that would otherwise interact with the cable. The experiment can then examine stimulation during a broader set of movements.



Montgomery et al. (2015), Fig. 1(a–c). Wireless-power preparation and miniaturized implanted light sources.



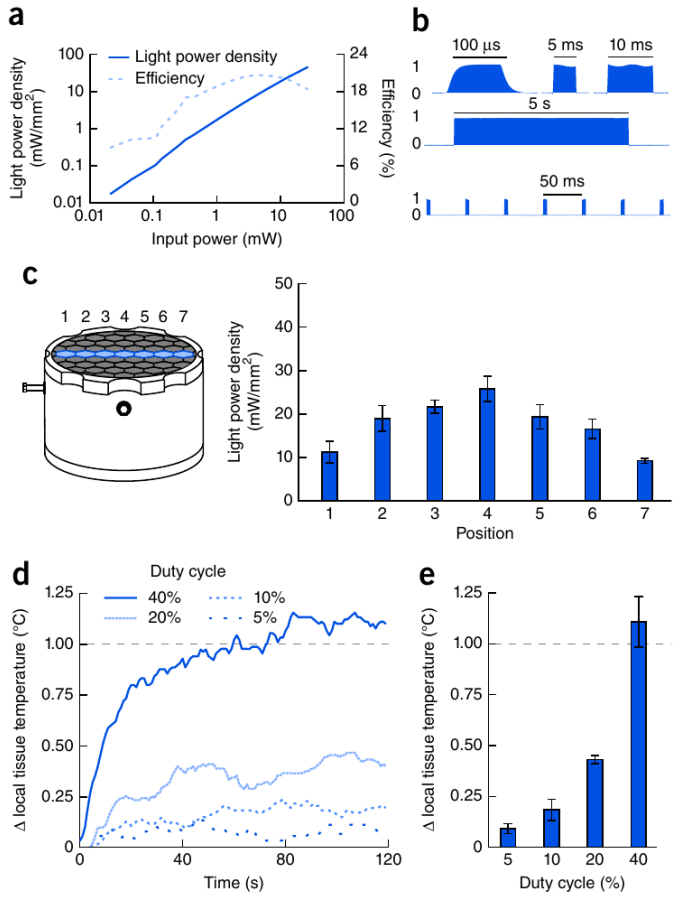
Montgomery et al. (2015), Supplementary Video 2. Wireless light delivery during running, tube exploration, and social interaction.

Wireless optical stimulation delivers pulsed energy

A light-emitting device uses electrical energy and produces heat. Montgomery and colleagues measured its light pulses and the temperature changes around it. Light output and heating are two consequences of powering the same device. Measuring both describes the optical input and the physical conditions around the implant.

Duty cycle is the fraction of time the light is on. Short pulses leave more time with the emitter off. Increasing duty cycle increases the average energy delivered. A pulse train has an intensity, pulse duration, and repetition rate. Together these determine how long the emitter operates during a given period of stimulation.

At 20 mW/mm² and duty cycles up to 20%, the measured temperature rise stayed below 1 °C. Pulse timing therefore changes both neural stimulation and the energy deposited near the tissue. The measurement concerns the tested stimulation conditions rather than every possible setting. Changing the timing changes how much energy is delivered between the start and end of an epoch.



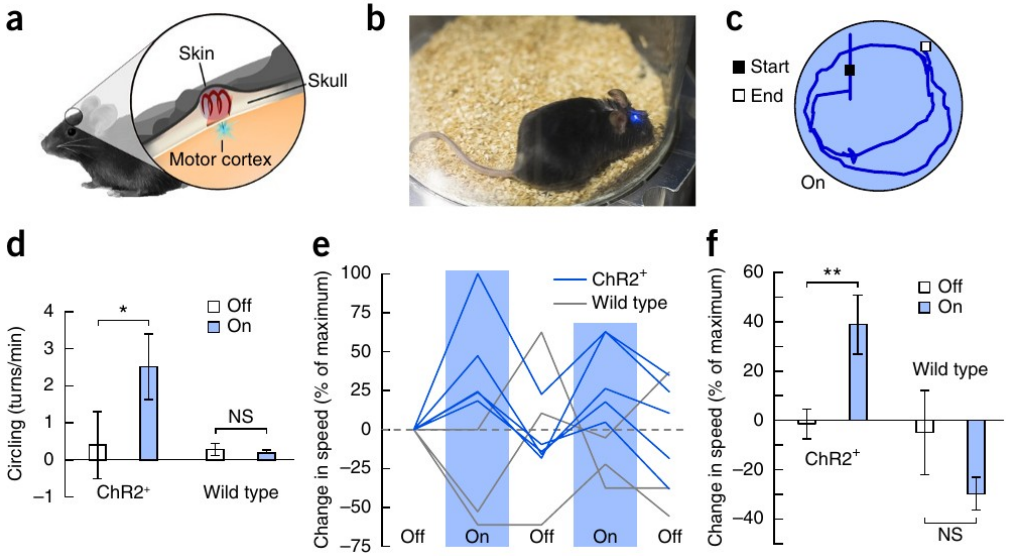
Montgomery et al. (2015), Fig. 2(a–e). LED power, pulse fidelity, position dependence and tissue heating.

Wireless cortical stimulation changes turning behavior

The **premotor cortex** helps organize movement. Montgomery and colleagues placed a wireless LED near the right premotor cortex of mice expressing ChR2. Selecting a movement-related target made turning a relevant outcome to measure. The implant provided an optical input while video recorded the mouse's trajectory through the arena.

During stimulation, 5-millisecond pulses at 20 Hz increased circling from about 0.4 to 2.5 turns per minute. Average movement speed also increased by about 40%. A turn count describes direction changes, while speed measures how quickly the mouse travels. Both changed during the stimulation periods, giving different measurements of the motor response.

Mice without the light-sensitive channel did not show the corresponding change. The light acted through the expressed channel, increasing neural activity that altered the animal's movement. The comparison separates an effect of illumination from an effect mediated by ChR2. The behavior changes when light can open the introduced channel in the targeted tissue.



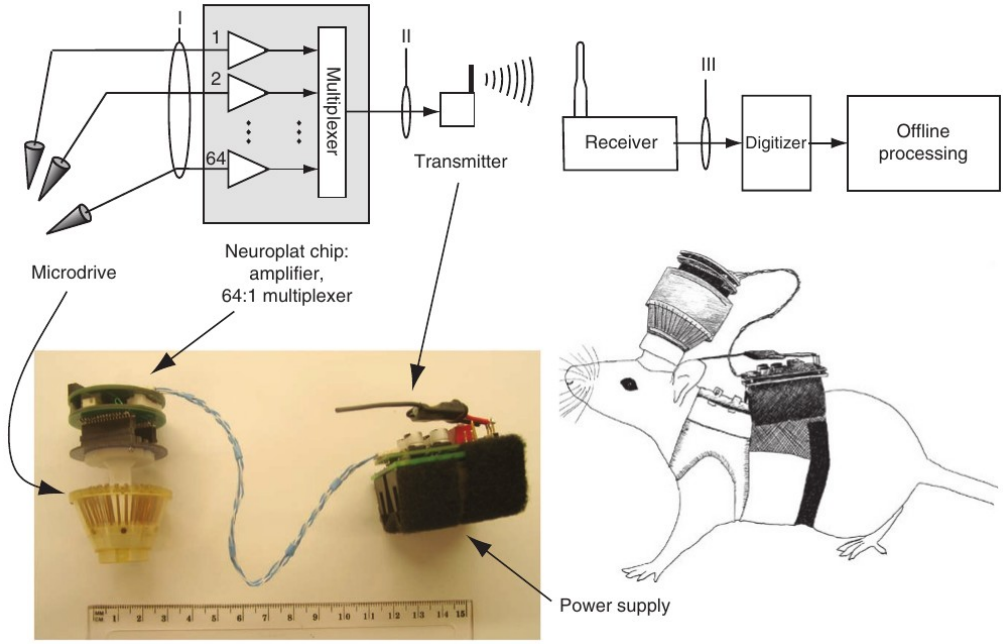
Montgomery et al. (2015), Fig. 3(a–f). Implanted premotor-cortex stimulation and circling behavior.

Wireless telemetry removes the external recording tether

Neural telemetry sends recorded brain signals to a remote receiver. Szuts and colleagues placed electrodes and an amplifier on a rat, with a radio transmitter carried in a harness. The electrodes measured activity on the animal, and radio transmission carried the data elsewhere. Recording and storage could therefore occur at separate physical locations during movement.

The amplifier increased the small voltages measured near neurons. The transmitter sent these measurements to a computer. The rat had no recording cable running to the external receiver. Amplification makes a small neural voltage easier for the electronics to process. Transmission then preserves the measurements as a stream that the receiver can reconstruct and store.

The system recorded during outdoor movement and travel through tunnels. Its larger battery configuration operated for about six hours. Researchers could measure neural activity while the animal moved through these spaces. Longer operating time permits behavior to be sampled across several episodes.



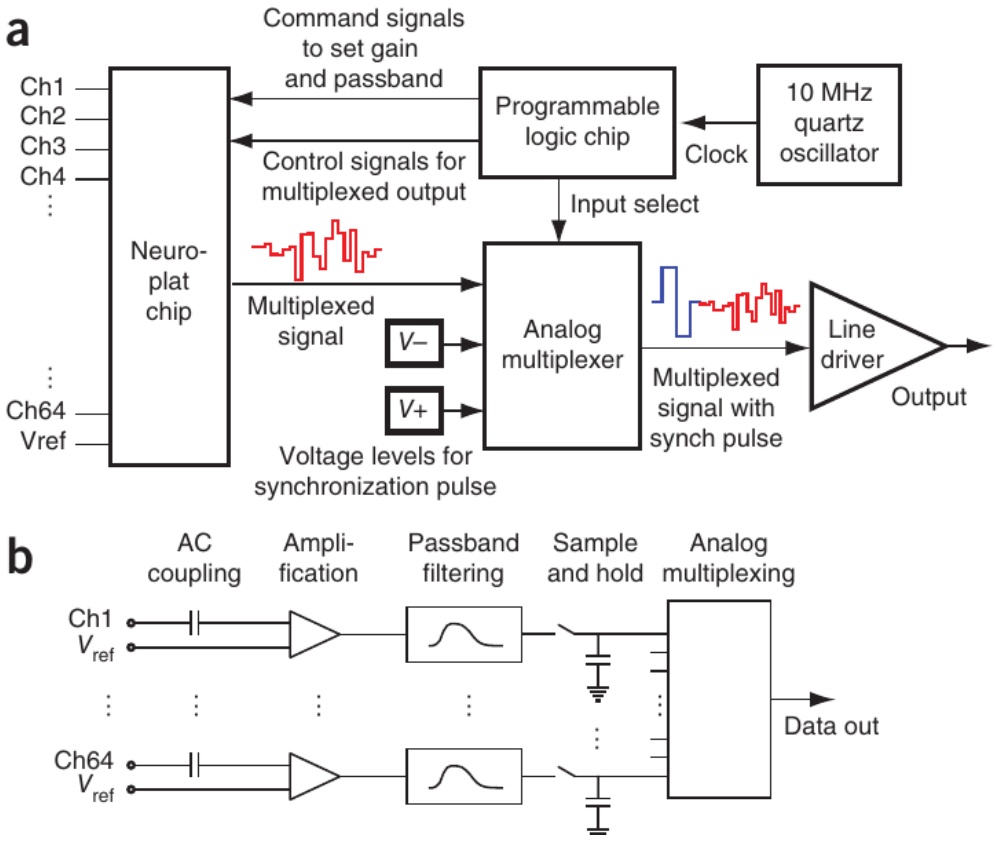
Szuts et al. (2011), Fig. 1. Published wireless headstage, harness and recording components.

Electrodes turn neural activity into a recorded signal

An **electrode** is a conducting contact that measures electrical changes. Near a neuron, it detects **extracellular signals**, small voltage changes outside the cell during neural activity. An extracellular recording samples nearby electrical changes rather than the entire brain at once. Electrode location therefore determines which local activity the system can detect.

A recording system takes repeated measurements. Its **sampling rate** is the number taken each second. Closely spaced samples capture brief spikes and preserve their timing. An action potential is brief, so infrequent sampling can miss parts of its waveform. Repeated measurements preserve changes that allow individual events to be recognized in the signal.

Wireless transmission sends those measurements to a receiver. Comparing spike times with the animal's movements identifies when a neuron was active during an action. The behavioral recording supplies the event times and positions for comparison. A neuron's firing can then be examined during approach, turning, travel, or another defined part of the movement.



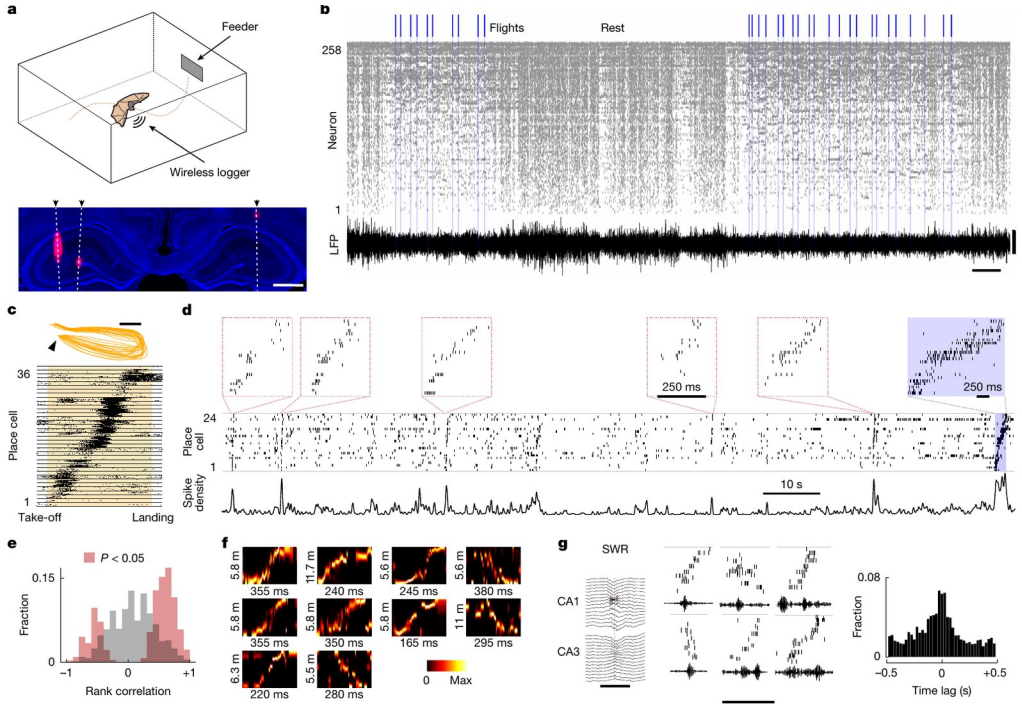
Szuts et al. (2011), Fig. 2(a–b). Sampling, multiplexing and signal conditioning in the original device.

Wireless recordings follow brain activity during bat flight

Forli and colleagues recorded neurons while bats flew freely. They studied the **hippocampus**, a brain region involved in memory and spatial representation. Wireless equipment allowed recording during flight. The animal moved while recording continued, so the brain signals were measured during the action itself. Flight supplied changes in position across both horizontal and vertical space.

Some neurons fired more strongly at particular locations. The researchers compared spike times with the bat's position along a flight path. A location-related response is identified by comparing activity across portions of the route. Repeating a trajectory allows the researcher to examine whether the spatial pattern recurs.

Different cells were active in different parts of the route. Their combined activity described spatial position while the bat moved through a three-dimensional environment. A group of cells can represent different portions of the same movement. Reading their activity in sequence describes how the representation changes as the bat progresses along the path.



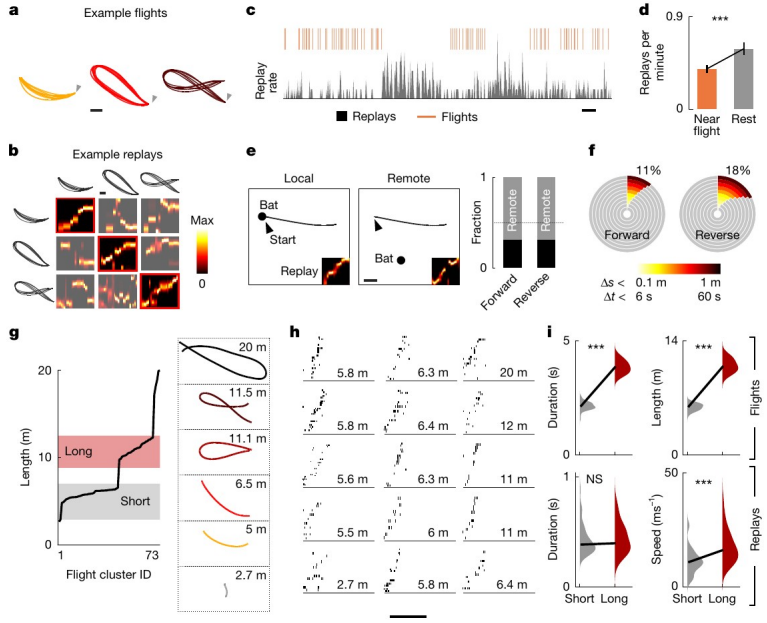
Forli et al. (2025), Fig. 1(a–j). Wireless Neuropixels recordings and flight-sequence replay.

Resting bats replay activity patterns from flight

Forli and colleagues compared hippocampal activity during flight and rest. **Replay** is a later sequence resembling the order of activity associated with an earlier experience. The comparison uses the order of activity across neurons, not only the total number of spikes. Similar ordering can identify a route-related sequence during a later period.

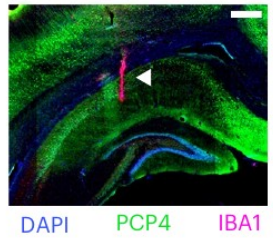
During some rest events, neurons repeated an order associated with a flight path. The later sequence occurred over a shorter time than the original movement. Compression means that the sequence occupies less time during rest. It does not mean the resting animal physically travels the same path at a faster speed.

Spatial experience therefore affected activity after the bat stopped flying. The hippocampus organized route-related sequences during both active navigation and subsequent rest. Rest activity provides a way to examine what happens after the immediate sensory and motor episode ends. The earlier experience can still organize the order in which neurons become active.



Forli et al. (2025), Fig. 2(a–i). Replay location, timing and scaling across flight trajectories.

d



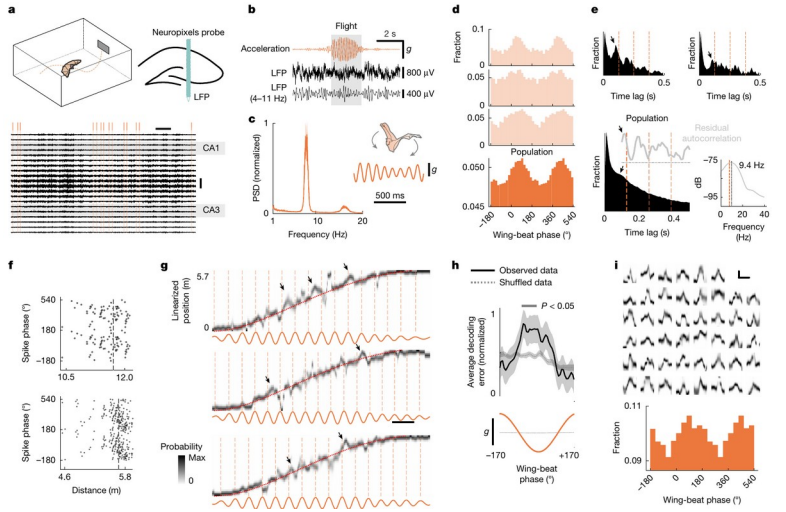
Snyder et al. (2024), Fig. 1(d). Recorded dorsal hippocampus with electrode tracks and a 500-µm scale bar.

Neural representations change across the flight rhythm

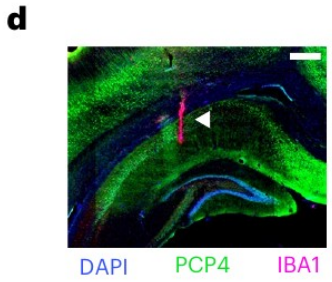
A **motor rhythm** is a repeating movement cycle. Successive wingbeats provide a rhythm during bat flight. Forli and colleagues aligned hippocampal recordings with phases of this cycle. Aligning recordings to the wingbeat identifies a cell's activity at comparable moments across cycles. Timing can therefore be measured relative to the animal's own ongoing movement.

Spatial representations changed across the wingbeat rhythm. Activity varied with the bat's location and with the timing of its movement. Location alone does not specify the whole pattern. The same spatial representation can vary over the phases of a repeated motor action as the animal continues flying.

A navigating brain processes information while the body moves. Recording the movement cycle and spike times together reveals when spatial signals occur during that action. Combining position, movement phase, and firing times gives separate measurements of space and timing. Their joint variation describes how neural activity is organized during active navigation.



Forli et al. (2025), Fig. 3(a–k). Representational sweeps and wing-beat phase during flight.



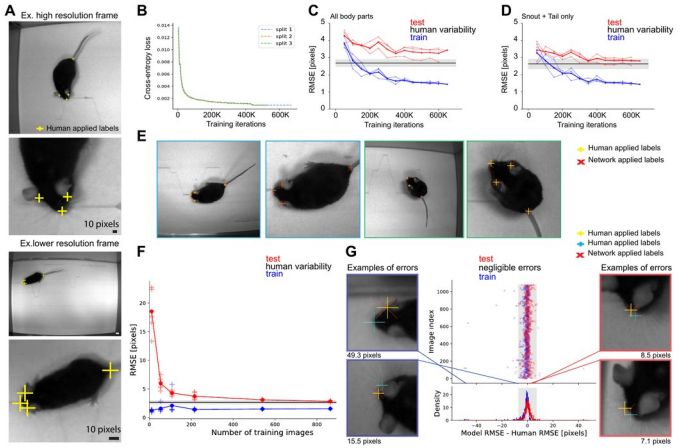
Snyder et al. (2024), Fig. 1(d). Recorded dorsal hippocampus with electrode tracks and a 500-µm scale bar.

Body landmarks make movement measurable

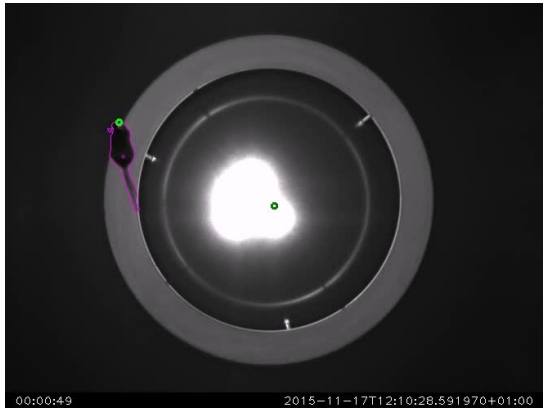
An **anatomical landmark** is a recognizable body feature used for measurement. Snout position describes head movement; wrist and digit positions describe a reach. Each landmark measures a particular part of an action. A landmark must be defined consistently across frames. Marking the same anatomical point allows its changing position to describe movement rather than changes in the labeling rule.

A researcher marks the landmark in selected video frames. A trained computer system then locates it in more frames, producing a record of its changing position. A sequence of positions records where the feature was at successive times. Comparing several features also describes orientation, such as the direction of the head relative to the body.

The Stowers demonstration marks a mouse's head in green and its body outline and center in magenta. These measurements allow the computer to update a visual scene according to the animal's position. The detected head location determines the viewpoint used for rendering. A computer can therefore use a behavioral measurement as an input to the next visual scene shown to the animal.



Mathis et al. (2018), Fig. 2A–G (author preprint). Tracking precision and training examples.



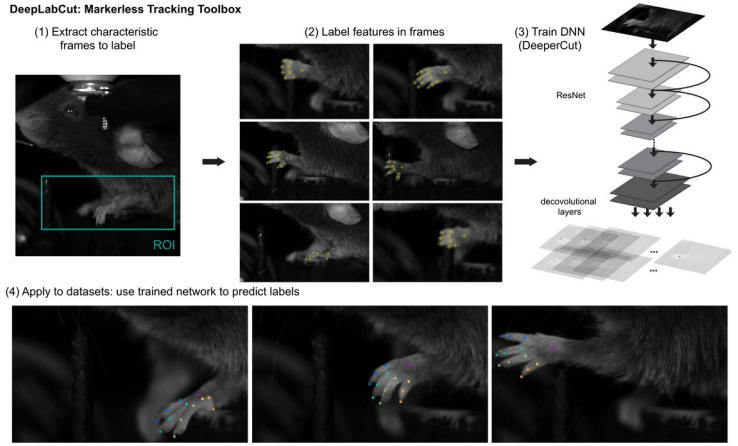
Landmark context: Stowers et al. (2017), Supplementary Video 9. Real-time mouse-head tracking.

Computer vision measures the position of body parts

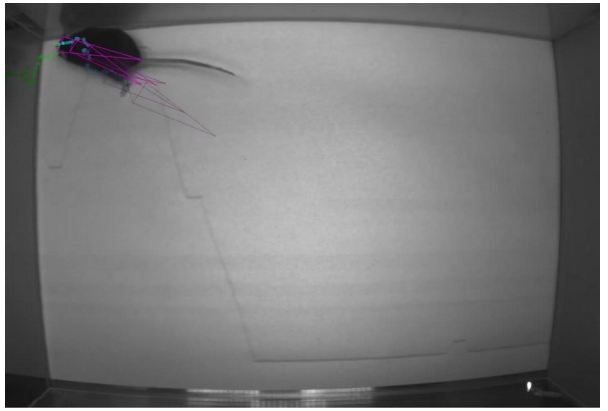
Computer vision extracts information from images. **Pose estimation** locates body parts such as a snout, wrist, or digit. Repeating this over video frames measures movement. A position estimate identifies where a feature lies within each image. Changes across successive frames describe the movement of that feature instead of requiring a score for every frame.

Mathis and colleagues developed DeepLabCut. A researcher marks selected body parts in example frames, and the trained system locates them in additional frames. The marked examples provide the information used during training. Once trained, the detector can apply those visual patterns to a much longer sequence of recorded frames.

The method tracked mice following odor trails, flies moving in chambers, and mice reaching for food. Choosing different landmarks allowed the same software to measure different actions. Different tasks require different features. Tracking the snout measures sampling movements, while tracking a wrist or digit measures a component of a skilled reach.



Mathis et al. (2018), Fig. 1 (author preprint). Labeling, training, and markerless pose estimation.



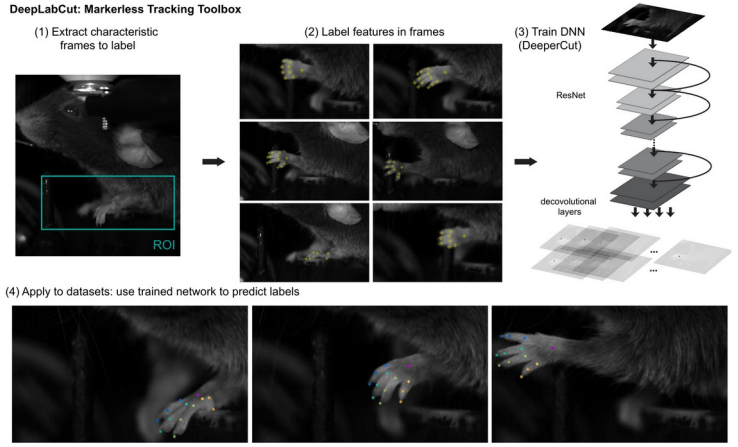
Mathis et al. (2018), Supplementary Video 1. Markerless tracking of a mouse following an odor trail.

Transfer learning adapts visual features to a new task

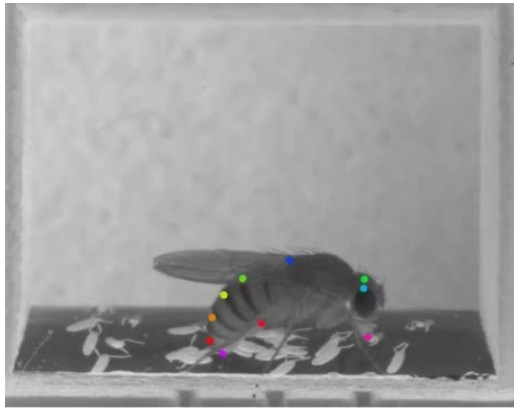
Transfer learning adapts information learned for one computational task to another. DeepLabCut started with features learned from many images, then trained the system to locate selected body parts. Features learned from a large image collection can help identify edges and shapes in a new task. Task-specific training then adjusts the detector to the selected anatomy.

An **artificial neural network** is a computer system whose adjustable processing steps transform an input into an output. Here, an image enters and predicted body-part positions come out. Training changes the parameters used to generate the output. The computer is learning a mapping from image information to labeled positions, rather than learning the animal's behavioral goal.

The software tracked both mouse and fly features. Reusing learned visual features reduced the amount of task-specific labeling needed. Researchers selected the anatomy they wanted to measure. The researcher still determines which landmarks matter for the question. A well-defined feature turns the detector's output into a measurement that can describe the selected behavior.



Mathis et al. (2018), Fig. 1 (author preprint). Labeling, training, and markerless pose estimation.



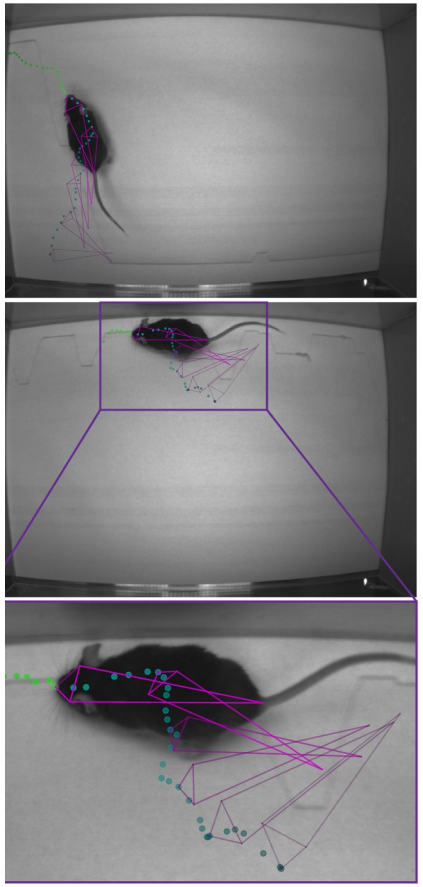
Mathis et al. (2018), Supplementary Video 2. Markerless tracking of fly body parts.

Tracking resolves head movements during odor following

A mouse following an odor trail moves its snout across the trail. Mathis and colleagues tracked the snout, ears, and tail base to measure head position and body orientation. Tracking several body parts distinguishes a moving head from movement of the whole body. Ear and tail positions provide reference points for interpreting the snout's direction.

A **trajectory** is a series of positions over time. The snout trajectory records the side-to-side movement as the mouse samples the trail. The trail supplies a position against which the trajectory can be compared. Side-to-side snout movement describes where the mouse samples relative to that external source of odor.

The same method also followed fly body parts and the **proboscis**, its feeding structure. Tracking selected features separates a head movement or mouthpart extension from movement of the whole animal. A whole-body position would miss some of these smaller actions. Selecting a mouthpart or a head feature gives a measurement at the scale required by the behavioral question.



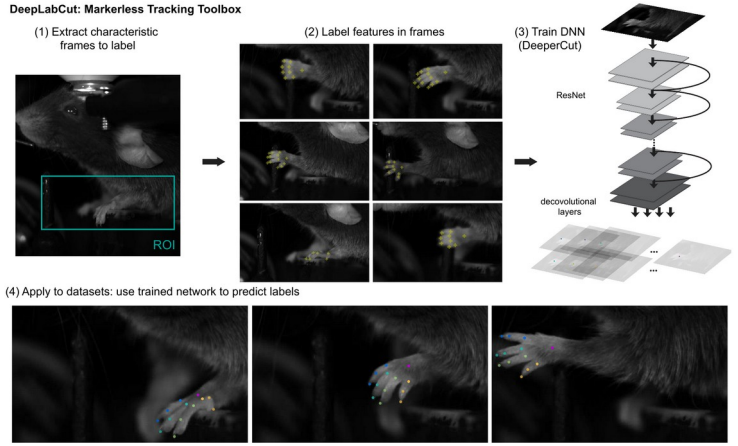
Mathis et al. (2018), Fig. 3 (author preprint). Snout and body trajectories during odor tracking.

Digit tracking resolves a skilled reaching movement

A reach involves the forelimb, wrist, and digits. Mathis and colleagues tracked selected landmarks as a mouse moved its paw toward a food target. Different joints move as the limb approaches the goal. Following more than one landmark describes the changing configuration rather than treating the entire hand as a single point.

Digit-joint positions changed as the paw approached and grasped. Several landmarks described the shape of the hand during the movement, rather than only its overall location. The wrist can move forward while individual digits change position relative to it. Measuring both distinguishes transport of the hand from the adjustments involved in grasping.

The measured anatomy determines the behavioral question. A snout trajectory describes sampling during odor navigation. Digit and wrist trajectories describe how the hand performs a skilled reach. The same video can provide several measurements when several landmarks are defined. These measurements describe different components of the action without requiring a separate camera for every feature.



Mathis et al. (2018), Fig. 1 (author preprint). Labeling, training, and markerless pose estimation.



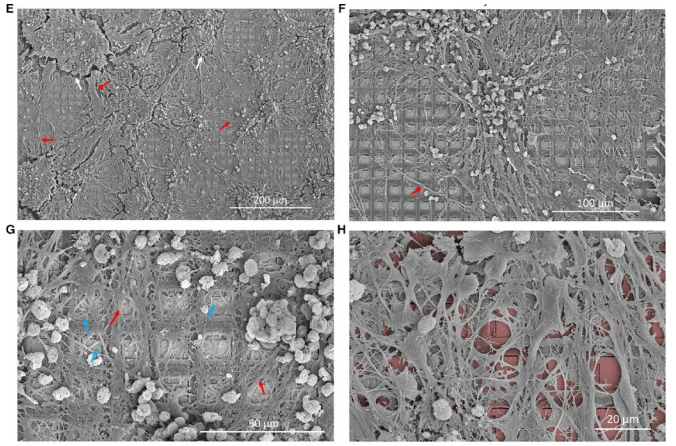
Mathis et al. (2018), Supplementary Video 3. Mouse digit and wrist tracking during a reaching movement.

Neurons communicate through branching processes

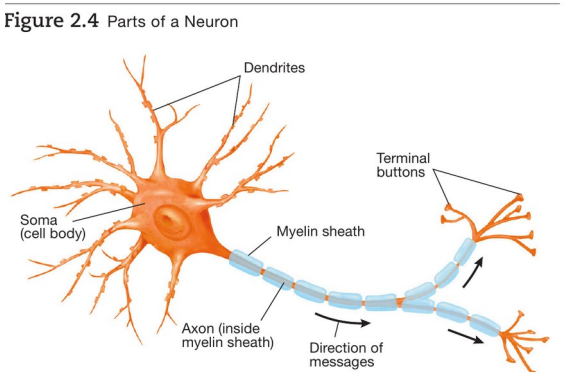
A neuron has a cell body and branching **processes**. **Dendrites** receive many incoming signals. An **axon** carries outgoing electrical activity toward other cells. A single cell can receive many inputs along its dendrites. Its axon distributes outgoing activity to terminals, where that activity can influence other cells in the network.

At a chemical synapse, the sending cell releases a neurotransmitter. The transmitter binds to **receptors**, proteins on the receiving cell, and changes that cell's response. Receptor activation determines the response at the receiving membrane. That response can change the cell's voltage and its tendency to generate its own outgoing action potentials.

Neuronal processes extend across the electrode array. An electrode records voltage changes nearby. Stimulation at one location can affect cells elsewhere as activity travels through the living network. The electrode does not need to be placed inside every cell to sample local activity. Multiple contacts across the surface allow signals from different locations to be compared over time.



Kagan et al. (2022), Fig. 2(E–H). Cultured neurons and processes span the electrode surface.



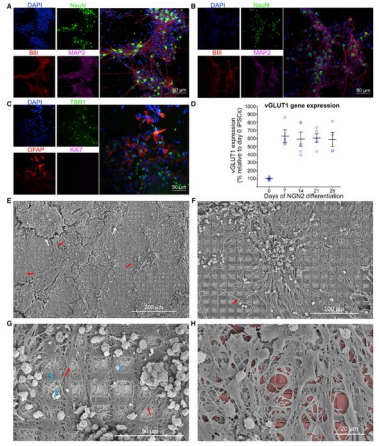
Carlson and Birkett (2022), Fig. 2.4, p. 40. Labeled cell body, dendrites, axon, and terminals.

Living cultures contain neurons and supporting cells

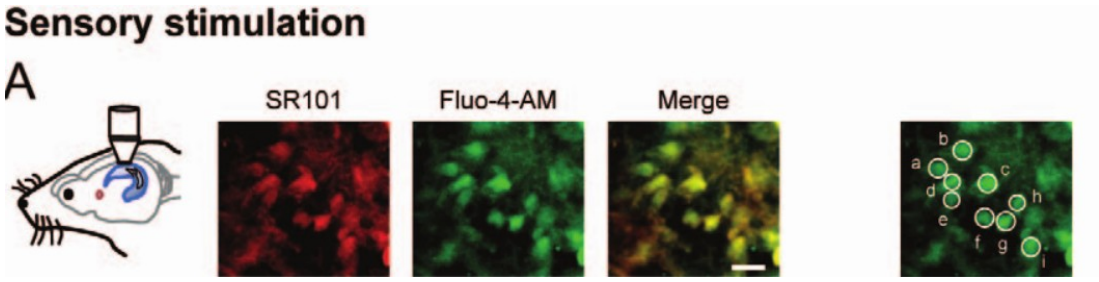
Kagan and colleagues grew neurons from mouse cortex and human **induced pluripotent stem cells**, adult cells reprogrammed to produce specialized cell types. The cells formed active cultures outside the body. Reprogramming changes the developmental possibilities of the source cells.

Cultures contain neuronal cell bodies and processes, plus supporting cells such as **astrocytes**. Astrocytes help maintain the chemical environment and supply metabolic support. The added micrograph identifies this cell type in a separate preparation. A supporting cell is not simply another kind of neuron.

The culture's neurons sent electrical signals through their processes. Electrodes recorded that activity and delivered stimulation. In DishBrain, the software used the signals to move a virtual Pong paddle. The array provided both a way to read activity and a way to deliver inputs. The task used those two directions of communication to make the tissue participate in a feedback cycle.



Kagan et al. (2022), Fig. 2(A–H). Neurons, neurites, astrocytes, and glutamatergic markers in cortical cultures.



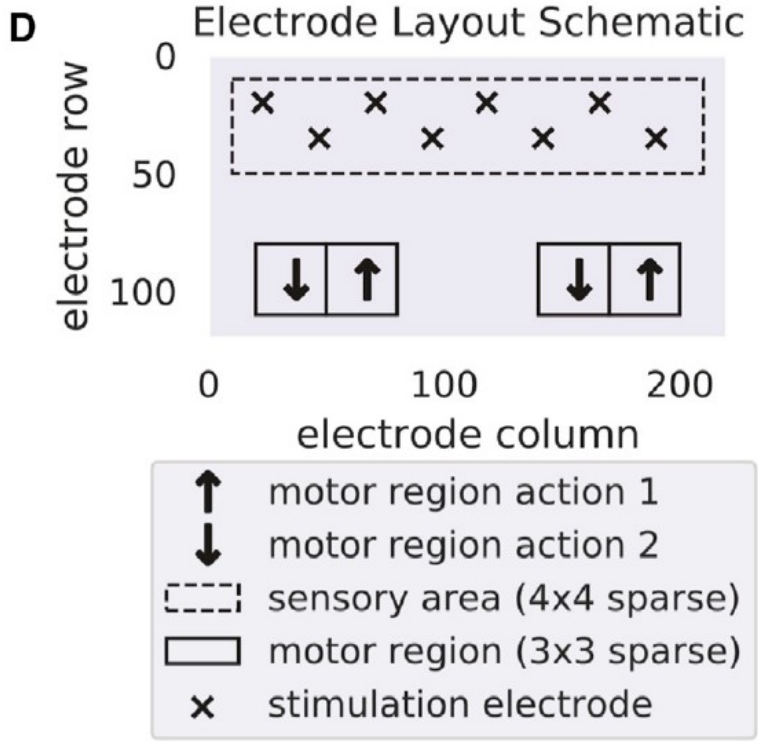
Navarrete et al. (2012), Fig. 1A. Fluorescently labeled astrocytes and calcium-indicator signal; original 20 µm scale bar.

Place and rate codes convey the virtual ball position

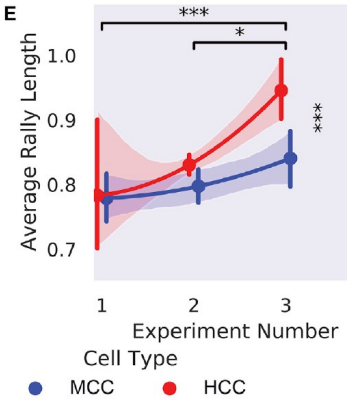
DishBrain converted the virtual ball's position into electrical pulses. Eight sensory electrodes delivered inputs. A **place code** uses the location of stimulation to represent a variable. A different active input site conveys a different ball position. The culture receives a spatial pattern of stimulation corresponding to the state of the simulated environment.

A **rate code** uses how often pulses occur. The refined protocol used rates from 4 to 40 Hz, giving the culture additional information about the ball. A higher rate supplies more pulses within the same period. Combining the active site with pulse frequency allows position and additional ball information to be carried together.

The neurons received timed electrical pulses rather than seeing a screen. Changing the encoding changed the information available for paddle control. Performance improved across the refined input designs. The biological preparation must respond to the supplied electrical pattern.



Kagan et al. (2022), Fig. 4(D). Electrode positions assigned to sensory and motor functions.



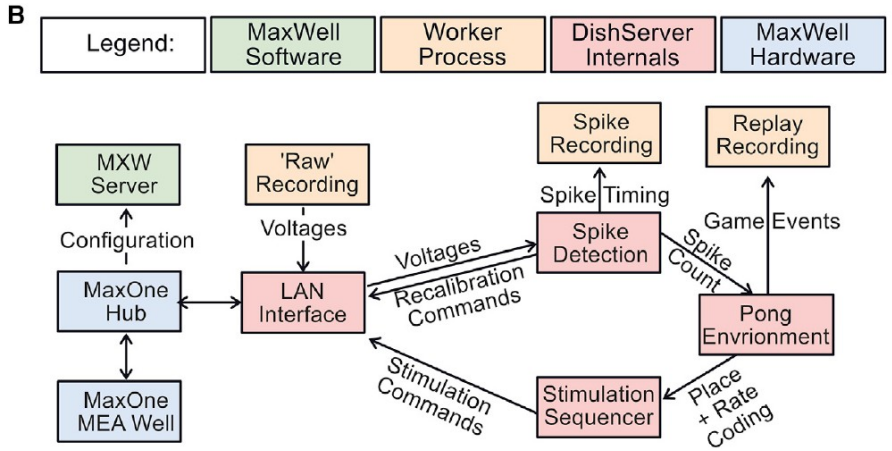
Kagan et al. (2022), Fig. 4(E). Gameplay performance changes across sensory-encoding designs.

An action determines the next input in a closed loop

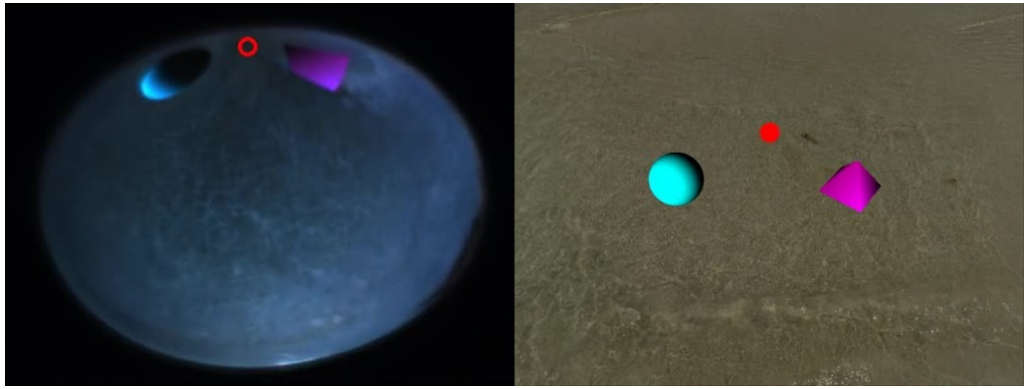
Feedback means that an action's result changes what happens next. During navigation, movement changes the visual scene. The new scene then guides another movement. The new visual input is a consequence of the earlier movement. Another action can then change the scene again, producing an ongoing cycle rather than one isolated stimulus and response.

In DishBrain, the neurons move the paddle. Whether the ball is hit determines the electrical signal sent back to them. Their activity changes the next input they receive. The culture's output therefore affects both the paddle and its later input. A different movement can cause a different game outcome and a different pattern of electrical stimulation.

A **closed loop** repeats this cycle of output and input. In the virtual-aquarium example, a fish's movement changes the displayed scene. In the culture, neural output changes the game and its subsequent stimulation. The input would follow a predetermined sequence in an open-loop arrangement. Here, the next event depends on the system's current output, making its own activity part of what controls subsequent stimulation.



Kagan et al. (2022), Fig. 4(B). Published data paths connecting neural activity and the game.



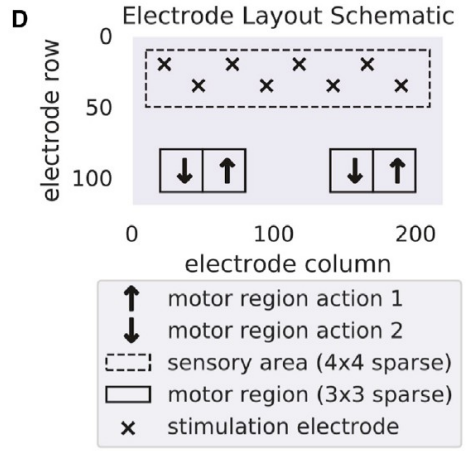
Feedback context: Stowers et al. (2017), Supplementary Video 3. Visual scenes around a freely swimming fish.

Motor-region activity determines paddle movement

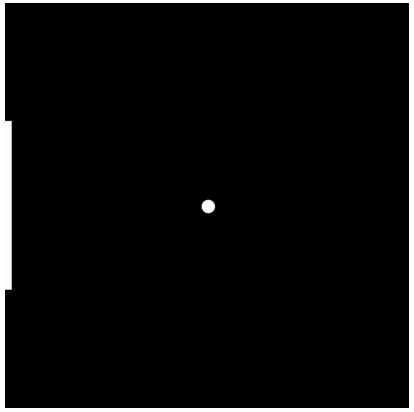
DishBrain assigned two electrode groups to paddle movement. Activity in one moved the paddle up; activity in the other moved it down. The movement command depends on which assigned group is more active. Two patterns containing spikes can therefore produce opposite paddle movements when their regional activity differs.

These **motor regions** were software-defined areas of the array, not an anatomical motor cortex. The computer compared their recorded activity to choose the paddle's movement. The label specifies how the computer uses a measured signal. It does not imply that the culture developed an anatomical region equivalent to the animal's normal movement-control cortex.

A hit required appropriately timed movement toward the ball. Spiking somewhere in the culture was not sufficient for effective control. The balance of activity in the assigned regions determined the action. The output must move the paddle to the required location before the ball arrives. The computer mapping makes differences in the timing and distribution of spikes behaviorally consequential within the game.



Kagan et al. (2022), Fig. 4(D). Electrode positions assigned to sensory and motor functions.



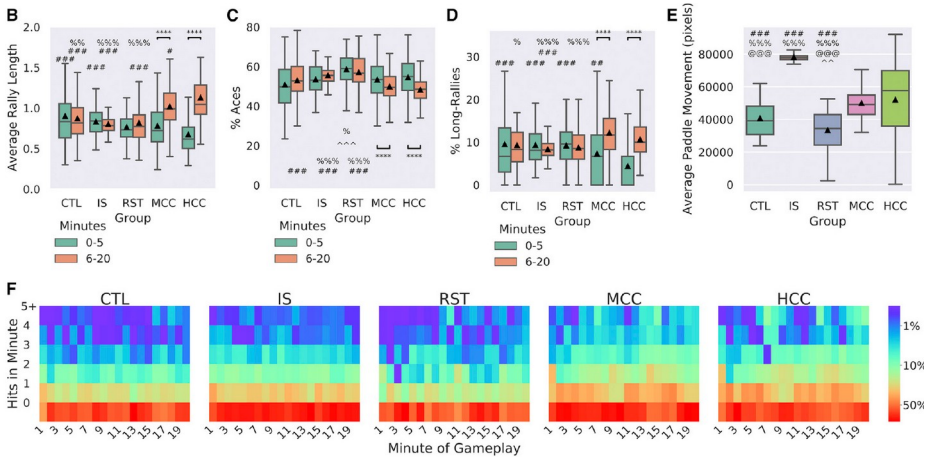
Kagan et al. (2022), Video S1. Neural control of a Pong paddle.

Mouse and human cultures improve during gameplay

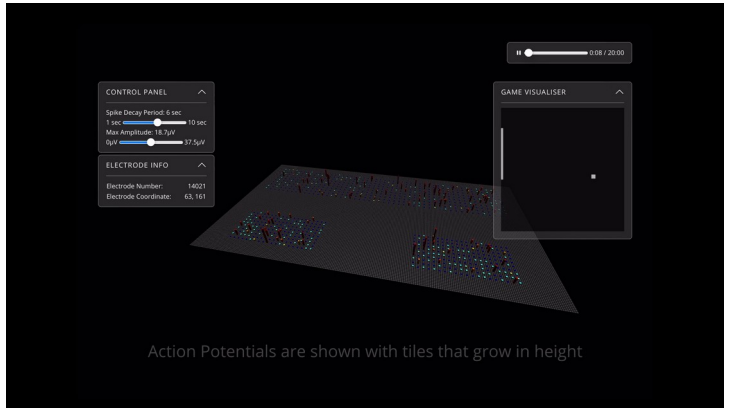
Average rally length counts how many hits occur before a miss. Kagan and colleagues compared the first five minutes of a session with minutes 6–20. A longer rally means the paddle intercepted the ball repeatedly before failure. This measurement captures sustained control rather than success on only the first approach of the ball.

Mouse and human cultures increased rally length during feedback-enabled gameplay. They also reduced immediate misses, called **aces**, and produced longer rallies. An ace ends the exchange before a successful return. Reducing aces and extending rallies describe two ways that performance became more effective over the course of interaction.

Improvement depended on participation in the task. The electrodes recorded neural output, the output moved the paddle, and the result changed the next stimulation. Repeated feedback allowed activity and performance to change during play. Each cycle presents another opportunity for activity to alter its own consequence.



Kagan et al. (2022), Fig. 5(B–F). Gameplay improvement, paddle movement, and hit distributions in experimental and control groups.



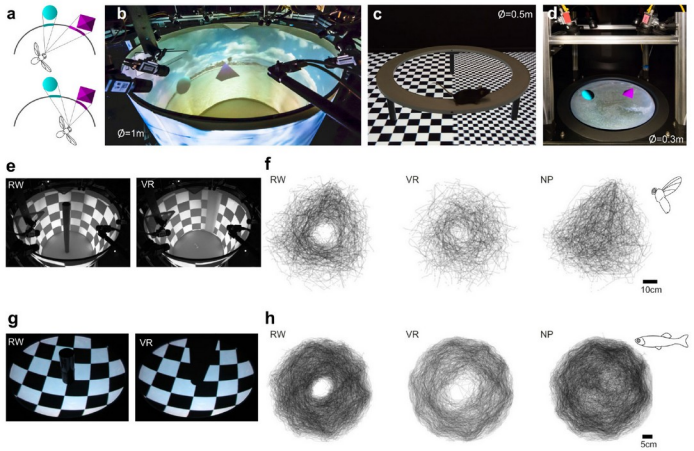
Kagan et al. (2022), Video S2. SpikeStream activity and Pong control.

A virtual world changes when the animal moves

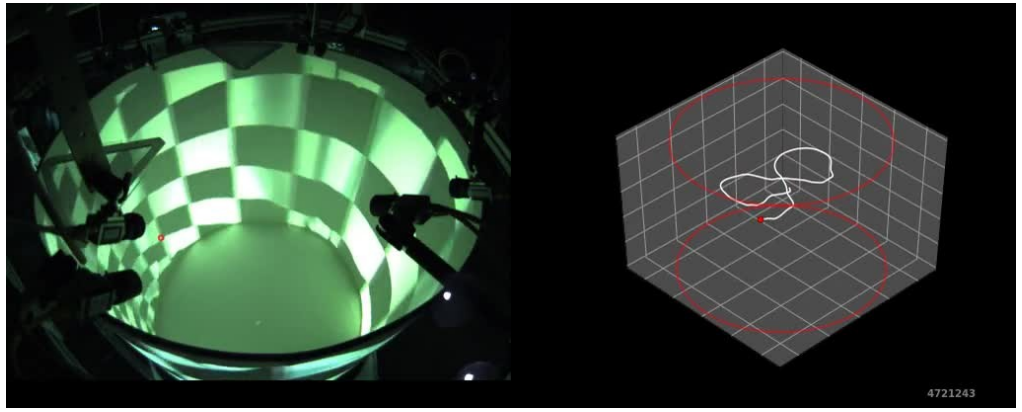
Stowers and colleagues measured an animal's position and used it to update a virtual scene. Moving through the arena changed the visual input delivered on surrounding displays. The animal's position served as an input to the computer. The resulting display then became a sensory input to the animal, allowing its action to change what it saw next.

Flies produce an **optomotor response**, turning in response to moving visual patterns. Rotating the virtual scene recruited this response and influenced free-flight direction. A fly can turn as the surrounding visual pattern moves. The researchers adjusted that pattern during flight, using an existing sensory-motor response to influence the next turn.

The researchers guided a fly along a figure-eight path using visual feedback. Measuring movement, changing the scene, and measuring the next movement made sensory guidance experimentally controllable. The imposed path emerged through repeated updates while the fly moved. The visual scene controlled the response through feedback rather than by physically forcing the body along the route.



Stowers et al. (2017), Fig. 1(a–h). Free-moving arenas and responses to real and virtual posts.



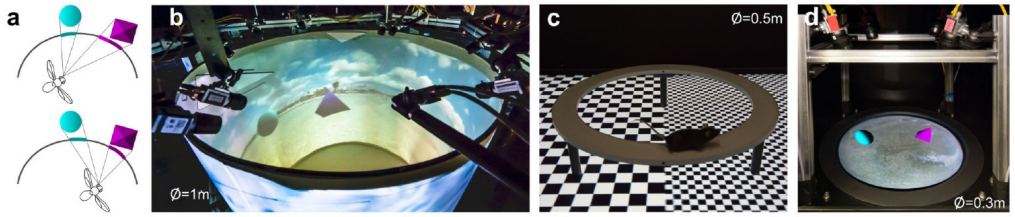
Stowers et al. (2017), Supplementary Video 10. Visual feedback guides free-flight turning.

Head position determines the animal’s visual perspective

An object’s appearance changes as an observer moves. Stowers and colleagues tracked position so a virtual object could be displayed from the observer’s current viewpoint. A nearby object changes apparent direction as the observer changes location. The system must account for that movement to maintain the object’s position within the simulated environment.

Perspective describes how the scene appears from that location. Multiple screens updated together, keeping a virtual object in a consistent place in the simulated world. Each screen supplies part of the view from the same tracked position. Coordinating their images avoids treating the animal as though it remained at one fixed location.

Movement then supplied information about spatial layout. The observer received a different image at a different position, as occurs with real objects. The demonstration makes this viewpoint-dependent update visible. These image changes provide information about distance and spatial arrangement. Without updating for movement, the display would supply a different visual experience from the intended virtual scene.



Stowers et al. (2017), Fig. 1(a–d). Perspective rendering and flight, mouse, and aquatic apparatus.



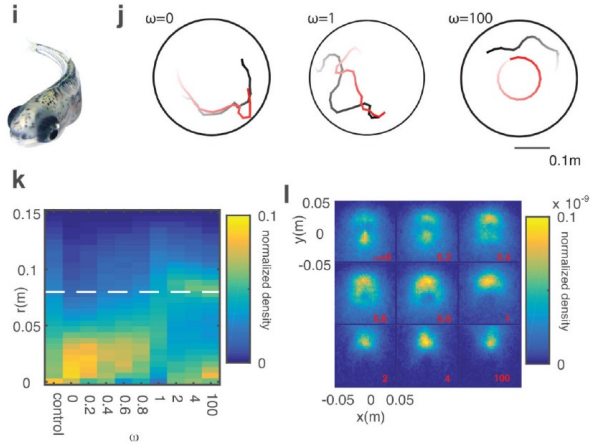
Perspective context: Stowers et al. (2017), Supplementary Video 1. Updating a scene for a moving observer.

Real and virtual fish influence each other

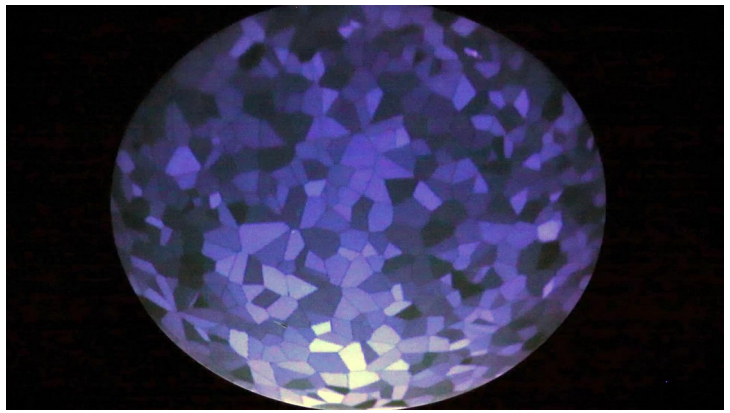
Stowers and colleagues displayed computer-generated partners to real fish. A **virtual partner** has an appearance and movement that researchers can control. Its appearance can resemble another fish while its movement follows an adjustable rule. This allows researchers to vary the social visual input without changing the real animal's physical tank.

The real fish's position changed the virtual fish's movement. The moving virtual fish could then influence the real animal. The two repeatedly altered one another's visual environment. The next movement on either side depends on the latest measured position. The interaction therefore continues through repeated updates rather than a fixed prerecorded swimming sequence.

Researchers varied the simulated partner's tendency to follow. Measuring the real fish's response tested how another apparent animal's movement influenced swimming during a social interaction. A stronger following tendency changes how much the virtual partner responds to the real fish. Measuring swimming under these settings tests the effect of that responsive visual behavior.



Stowers et al. (2017), Fig. 5(i–l). Real–virtual fish trajectories and effects of social coupling.



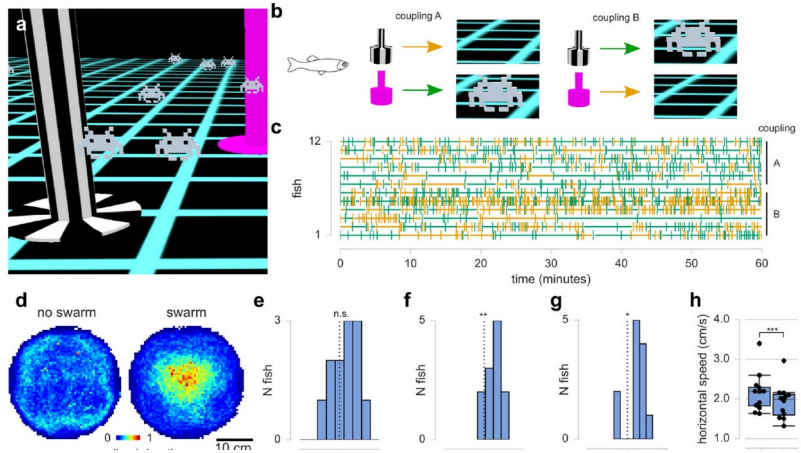
Stowers et al. (2017), Supplementary Video 14. Reciprocal movement of real and virtual fish.

Portals change a fish’s world without moving its body

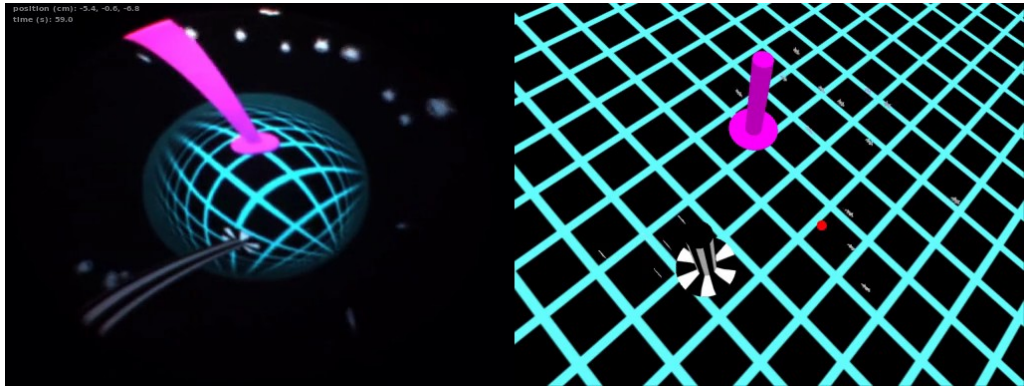
A **portal** was a region that changed the displayed world when a zebrafish entered it. The fish stayed in the same tank while the visual scene switched. The portal defines an action that triggers a consequence. Crossing its boundary changes the sensory surroundings, making entry choices distinguishable from movement within the newly displayed scene.

One scene contained a moving swarm of simulated agents. Fish stayed longer and swam more slowly in that scene. Their behavior changed after entering the environment. Longer residence concerns behavior after the scene becomes available. Slower swimming is another response within that environment, and can be measured separately from the initial portal entry.

Entrance choices did not favor the swarm outcome, although fish preferred magenta portals. Staying longer in a scene and choosing the entrance that leads to it are different behaviors. The task measured both. A preference for a portal’s appearance can differ from a preference for its outcome. Recording entry and residence separately keeps these two kinds of choice distinguishable.



Stowers et al. (2017), Fig. 5(a–h). Portal choices, scene occupancy, and swimming with a virtual swarm.



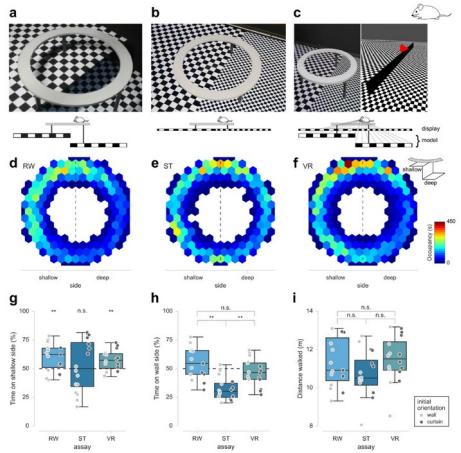
Stowers et al. (2017), Supplementary Video 13. Zebrafish portals and virtual swarms.

Freely moving mice avoid virtual depth

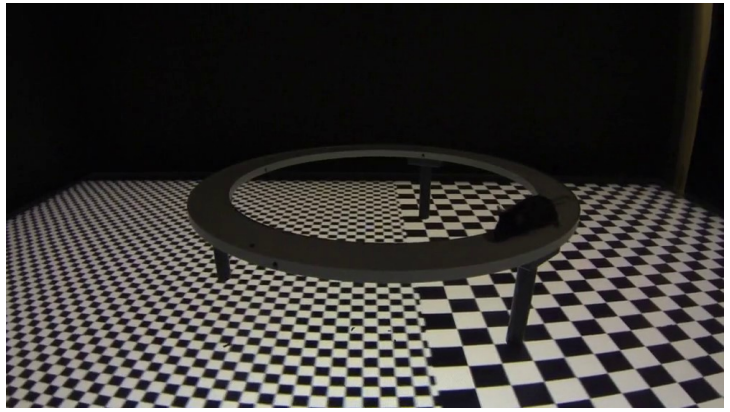
Stowers and colleagues placed mice on a platform above apparent shallow and deep spaces. Real surfaces were 20 and 40 centimetres below the track. Virtual scenes reproduced these views. The mouse's physical walking surface remained a platform. The manipulation changed what appeared below it, allowing apparent depth to be varied without giving the animal a different route to stand on.

Motion parallax is the change in an object's apparent position as the observer moves. Near and far objects shift differently. Head tracking allowed the displays to reproduce these differences. Updating for head position supplies the image changes normally produced by self-movement. Those changes can provide depth information beyond the size of a checkerboard element in a stationary image.

Mice preferred the shallow side with real surfaces and perspective-correct virtual scenes. A static display lacked the same preference. Movement-dependent visual changes helped guide where the mice walked. The comparison preserves visible texture while changing its movement-dependent behavior. Different walking choices therefore identify a contribution from the visual information supplied as the mouse moves.



Stowers et al. (2017), Fig. 2(a–i). Real, static, and perspective-correct depth conditions with occupancy measures.



Stowers et al. (2017), Supplementary Video 8. Mouse elevated-track depth assay.

Key takeaways

Behavior begins with neural signaling Sensory inputs change neuronal activity, and motor outputs produce actions; internal state and prior experience alter the response.

Plasticity changes what circuits can do Development, practice, and recovery after early injury alter neural organization and the skills remaining tissue can perform.

Mechanism and function answer different questions Mechanism explains the cells and signals producing an action; function describes what the action accomplishes.

Learning depends on consequences and feedback Rewarded actions become more effective, while a closed loop makes the next input depend on the system’s own output.

Neural tools test specific contributions Recording measures activity during behavior; optogenetic activation or inhibition changes selected cells so the resulting behavior can be measured.

Biological interfaces require defined inputs and outputs A fish controls wheels, a sea lion controls a cursor, and cultured neurons control a paddle through explicit mappings that researchers can test.

Carlson & Birkett (2022); Kliemann et al. (2019); Boyden et al. (2005); Mathis et al. (2018); Givon et al. (2022); Kagan et al. (2022)